

Section (6) of the **ORGANIZE** popup menu in project management contains functions of *data backup* and *data restoration*, such as packing or backing up projects, or the *e-mail function*. The e-mail function requires that a functioning e-mail client and associated profile of the user have been set up.

In section (7), a packed project (file extension *.elp) can be opened via the **UNPACK** command.

A packed project can also be opened in the project tree by double-clicking. EPLAN unpacks the project automatically, and then you can continue editing as usual.

Many of the functions accessed via the **ORGANIZE** popup menu can also be executed using the popup menu on the left (project tree).

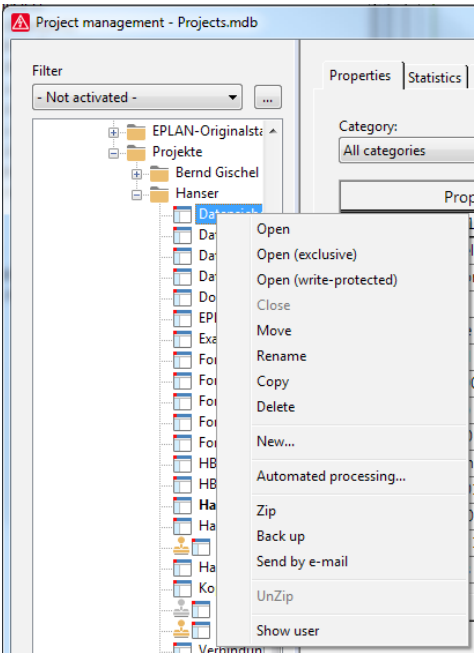


Fig. 7.140
Popup menu of the
right mouse button

7.6.2.6 Extras button

That leaves us only with the **EXTRAS** button. This one also includes a number of popup menu functions, which will be explained briefly.

These are functions of the project database, as well as functions for comparing or reading project information.

Section (1) of the **EXTRAS** popup menu contains functions to complete the project database, import completely new project directories, or generate a project list.

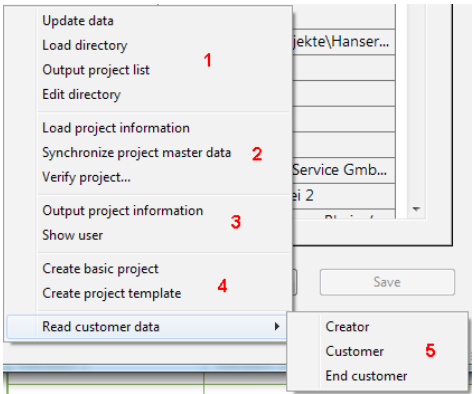


Fig. 7.141
Extras menu

Section (2) contains commands for reading out project information or, for example, for comparing a project with the set company parameters.

Among other things, section (3) provides an overview of which users are still in the selected EPLAN project.

Section (4) contains the familiar functions Create basic project and/or Create project template.

In section (5), customer data can be imported from parts management into project properties. Select the project in question and open the *Read customer data* menu item. Then, you

can click one of the menu items, *Creator*, *Customer* or *End customer*, to transfer the data from parts management (customer data) to project properties automatically.

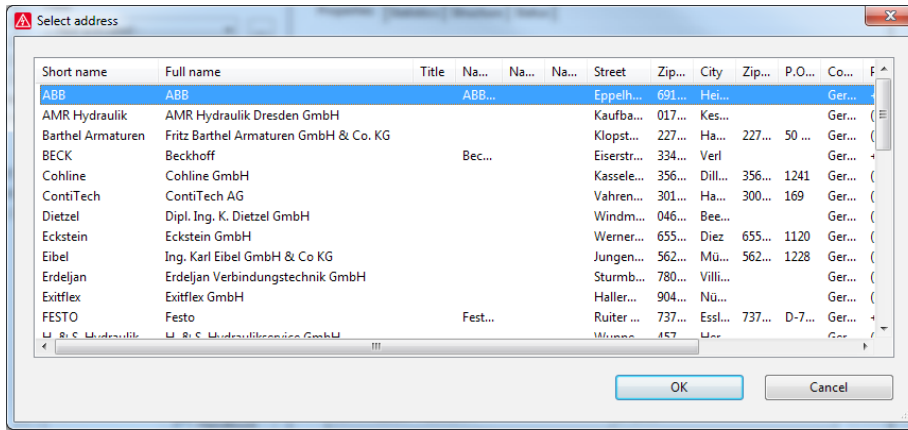


Fig. 7.142 Select address dialog

7.6.3 Project management filters

Over time, many projects are created. So as not to lose track, or to display only specific projects quickly, you can create filters in project management with a variety of possible report criteria via the project directories.

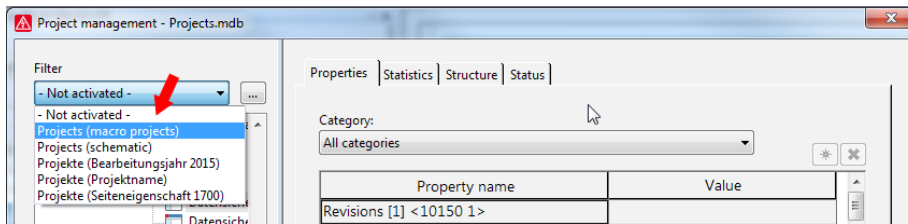


Fig. 7.143 Filter selection

You can select existing filter schemes in project management (upper left) using the *Filter* drop-down list. Of course, you can also create your own filters in the usual way.

If you have selected a filter scheme, but no projects are displayed, it is very likely that the criteria are not compatible with the set projects and/or project directories.

7.6.4 Create projects (project management)

Creating a new project is identical to creating one in the graphical editor. EPLAN works the same way here.

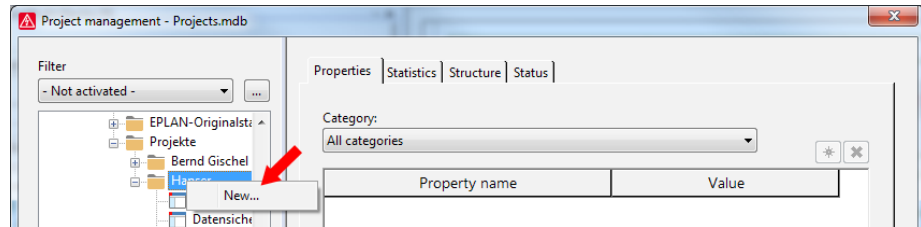


Fig. 7.144 Create new project in project management

In the left area of project management, you can use the popup menu or right mouse button to create a new project via the **NEW** menu entry and, in the subsequent dialog, on the basis of a project template or a basic project.

The further procedure, then, is identical.

7.6.4.1 Copy projects (project management)

Of course, apart from creating new projects, you can also copy projects in project management, and then adapt them to the required entries. EPLAN provides various options for copying a project.

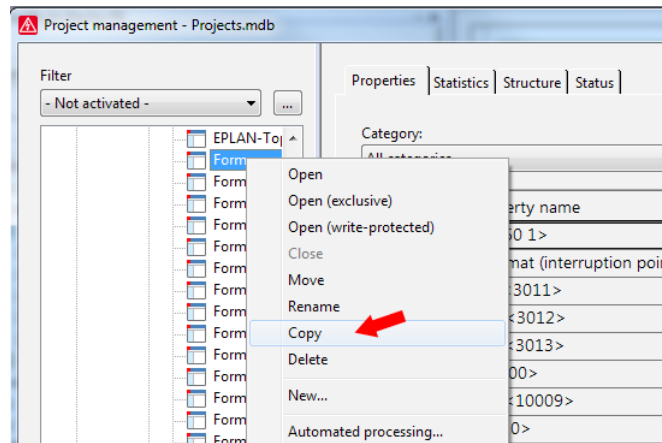


Fig. 7.145 Copy (project) popup menu

In the left area of the **PROJECT MANAGEMENT** dialog, select the project to be copied. Then, using the popup menu (right mouse button) or the **ORGANIZE** button, start the *Copy* function. EPLAN opens the **COPY PROJECT** dialog.

The **COPY PROJECT** dialog contains various options regarding how and to what extent the selected project is to be copied, and whether a specific creation date (project property 10021 creation date) and/or a specific creator (project property 10020 creator) is to be entered in the project copy automatically.

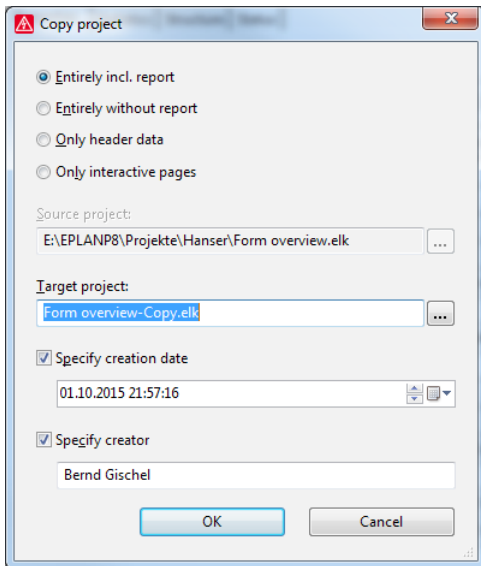


Fig. 7.146 Copy project dialog

You can also select and copy several projects in EPLAN. In the process, the **COPY PROJECT** dialog opens several times one after the other.

7.6.4.1.1 Entirely with report setting

The entire project, including header data, such as page structure and device structure, is copied together with all reports generated. This option generates an identical project copy.

7.6.4.1.2 Entirely without report setting

The entire project is copied, but without any existing (graphical) reports generated. After a project copy has been created, the reports must then be regenerated.



NOTE: With this copy option selected, any title page or cover sheet manually created will not be applied as a report either.

7.6.4.1.3 Only header data setting

The project is copied only with the existing header data, such as page structure and device structure. No existing pages or existing reports will be copied. The subsequent copy of the project is empty, which means that this project does not contain any pages.

7.6.4.1.4 Only logic pages setting

The project is copied with the existing header data, such as page structure and device structure, and also includes all logic schematic pages.

These include, for example, the following page types: External document, Graphic, P&I diagram, Schematic multi-line, Schematic single-line, Schematic (Fluid power), Panel layout, and Overview.

However, unplaced items or reports are not copied.



NOTE: Unplaced items are items that have not been placed in the project yet, that is, they have not been drawn yet and therefore have no graphical reference. They exist only in lists or in the navigators as devices – for example, unplaced terminal strip definitions.

7.6.4.2 Copy projects (Explorer)

Apart from the EPLAN copy function projects can also be copied “externally”. But this usually involves the mistake, for example, of only copying the project folder *.edb and forgetting about the project file *.elk for starting the project.

I strongly advise against it, because EPLAN cannot track such external copy actions automatically, nor does it register them in the project management database, for example. Projects should always be copied using the available EPLAN functions, and always within the EPLAN platform.

8

Export, import, print

EPLAN offers a range of export and import functions for the entire project or also individual pages. Export and import are relatively simple processes that are only carried out for the selected pages (or all pages) of a project.

An export can be launched via the **PAGE/EXPORT [EXPORT TYPE]** menu; an import via the **PAGE/IMPORT [IMPORT TYPE]** menu.

Export menu

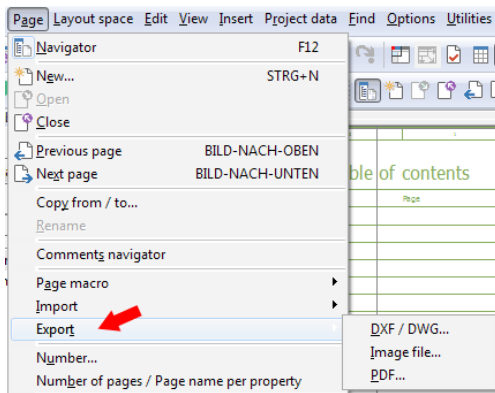


Fig. 8.1 Menu of miscellaneous export options

Import menu

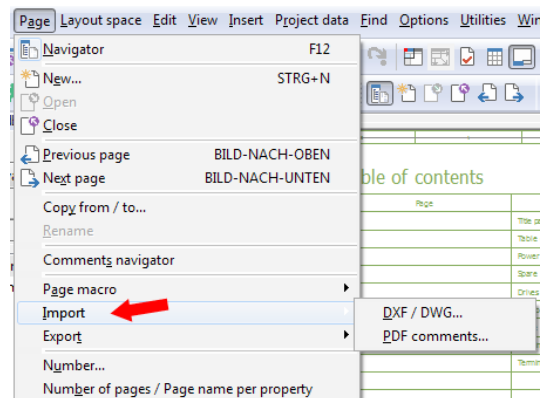


Fig. 8.2 Menu of miscellaneous import options



NOTE: Exported pages in an image format or DXF/DWG format do not contain any (electrical engineering) logic after the export.

This is why this export function is not suitable for exchanging projects. To exchange projects, you should always do a proper data backup (via the *Back up* or *Restore* project options).

NOTE: Pages can only be exported from one project at a time. Pages selected from different projects in the page navigator cannot be exported together.

The export and import options are then grayed out and cannot be selected.

8.1 Export and import of DXF/DWG files

Files in the DXF or DWG format have a neutral exchange format that EPLAN can use. However, after export or import, these files contain only purely graphical elements such as circles, lines, etc.

The DXF and/or DWG formats are widely used and popular in various CAD/CAE programs, for example, to represent machine layouts or similar elements. Therefore, they are ideal for exchanging non-program-specific drawings, etc., even if they are purely graphical.

8.1.1 Exporting DXF and DWG files

To export pages in DXF or DWG format, you first select the desired pages in the **page navigator** and then use the mouse to start the function from the **PAGE/EXPORT/DXF/DWG** menu (the ALT + S keyboard shortcut does not work when the page navigator is open). EPLAN opens the DXF/DWG export dialog and loads the **Source** field with the selected pages (this field cannot be edited by the user).

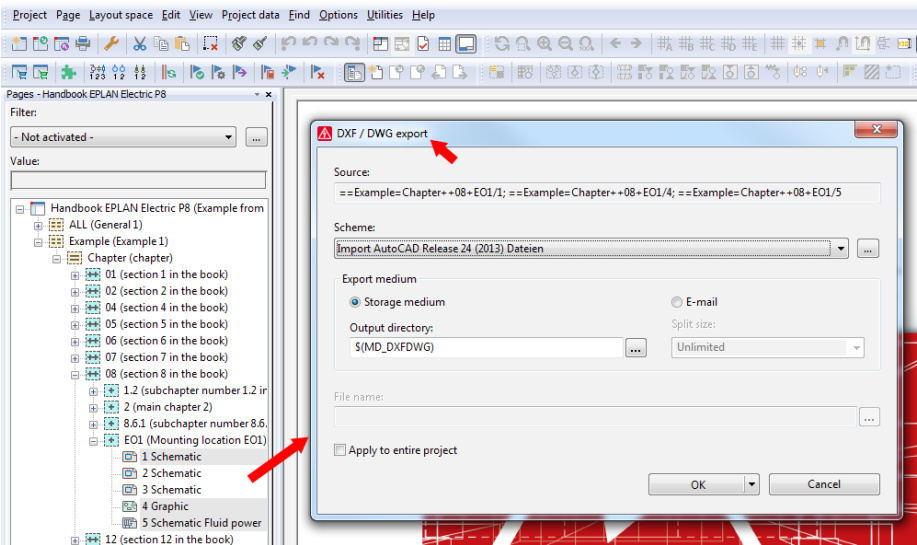


Fig. 8.3
Export pages to
DXF / DWG format
dialog

The **Export** scheme is more interesting and can be modified by the user. Existing schemes can be selected in the selection field here and then applied.

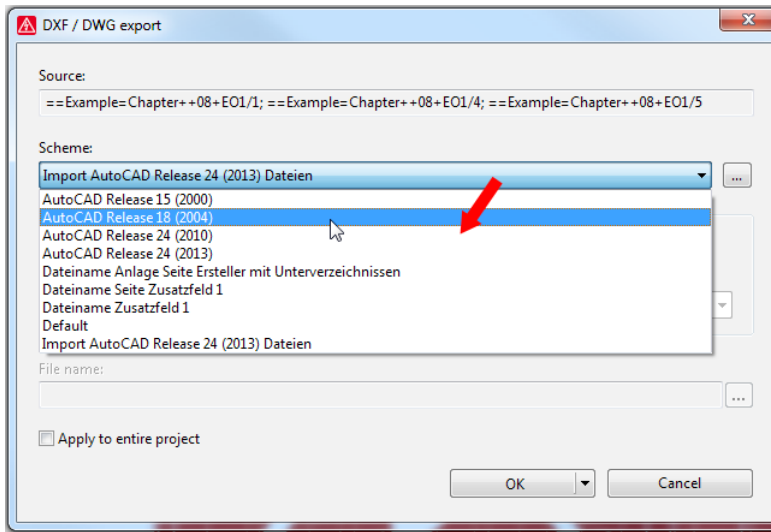


Fig. 8.4
Schemes can be adjusted in any manner

The [...] button opens the **Settings: DXF/DWG export and import** dialog, where the current scheme can be edited.

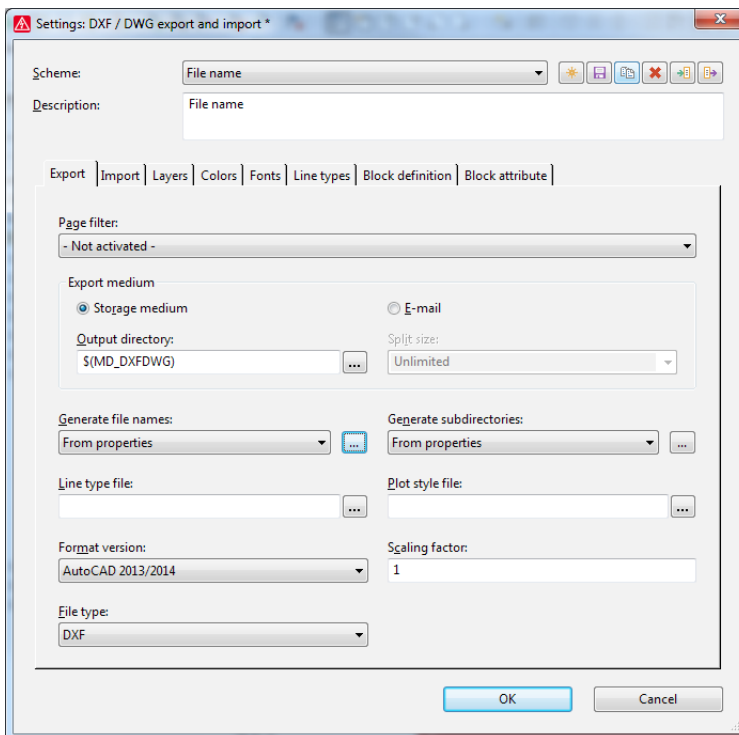


Fig. 8.5
Settings for exporting DXF / DWG files

These settings are not project-specific and are regarded as user settings. You can find them under **OPTIONS/SETTINGS/USER/INTERFACES/DXF/DWG EXPORT AND IMPORT**.

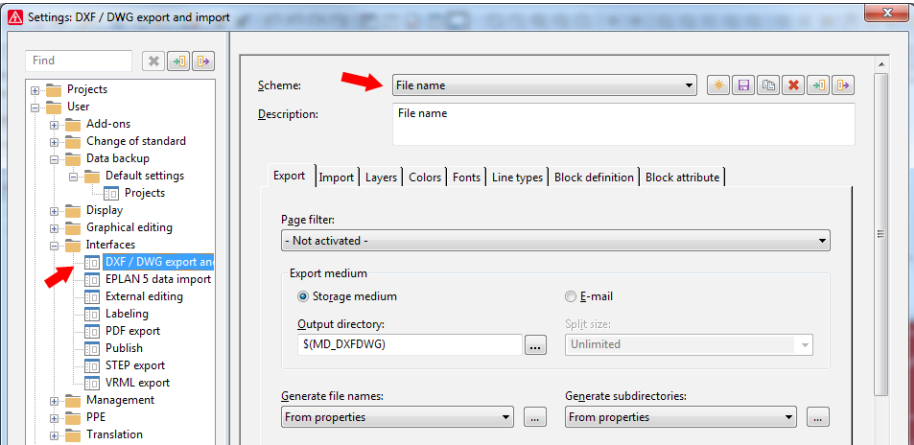


Fig. 8.6 Settings for exporting DXF / DWG files

DXF / DWG export and import settings

The **Settings: DXF/DWG export and import** dialog offers the user many options/parameters for influencing or controlling the export and import of DXF/DWG files.

A number of parameters are important for exporting pages in the DXF/DWG format. However, at this point you should create a new scheme (as usual, via the **NEW** button) and not edit the currently set scheme. An exception to this is when the scheme settings are not optimal. You can then change these in the current scheme and save the scheme under the same name.

Once a name and optional description for the scheme have been defined, you close the dialog by clicking **OK**. EPLAN enters the new scheme name into the **Scheme** selection field. The desired export settings can now be adjusted.

You can set the **Output directory** to a special directory for this specific project, instead of the usual global output directory for DXF/DWG files.

You should adjust the **Style version** to suit the target system. EPLAN provides a selection field with different entries and a certain number of different target systems.

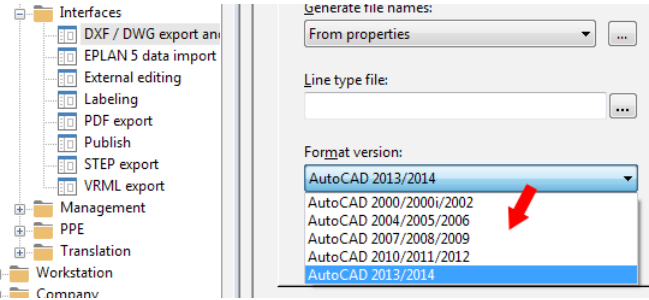


Fig. 8.7 Selection of formats

The **File type** can be set to either DXF or DWG. This setting is also made via a selection field and must be set to suit your requirements.

The **Scaling factor** controls enlargement or reduction of the page being output. The usual value is 1, i.e. the page is written to the DXF/DWG format on a 1:1 basis without enlargement or reduction. This entry should be left at the default value.

The *Generate file names* option allows the user to define the name of the generated DXF/DWG files. You can allow EPLAN to automatically generate the names using *consecutive numbers*, or you can use your own scheme to generate the file names individually from various page and/or project properties (*From properties* option).

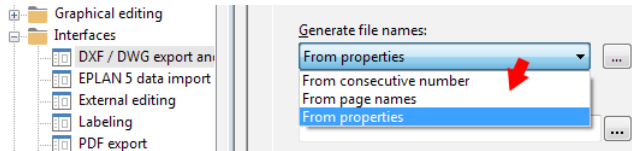


Fig. 8.8 File name options

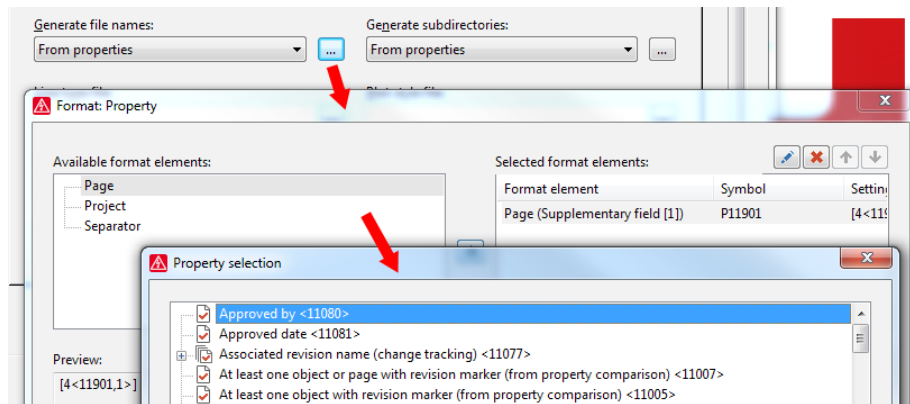


Fig. 8.9 Generating file names from properties (format elements)

Once all options or parameters have been set, you save the scheme via the graphical [SAVE] button. It is then always available and can also be used for other projects, as described previously.

It is also possible to output the DXF/DWG files to be generated in defined and fixed **sub-directories**. This would be practical, for example, when you want to structure the output in such a way that a subdirectory is generated for each higher-level function, with all pages belonging to the higher-level function being output there.

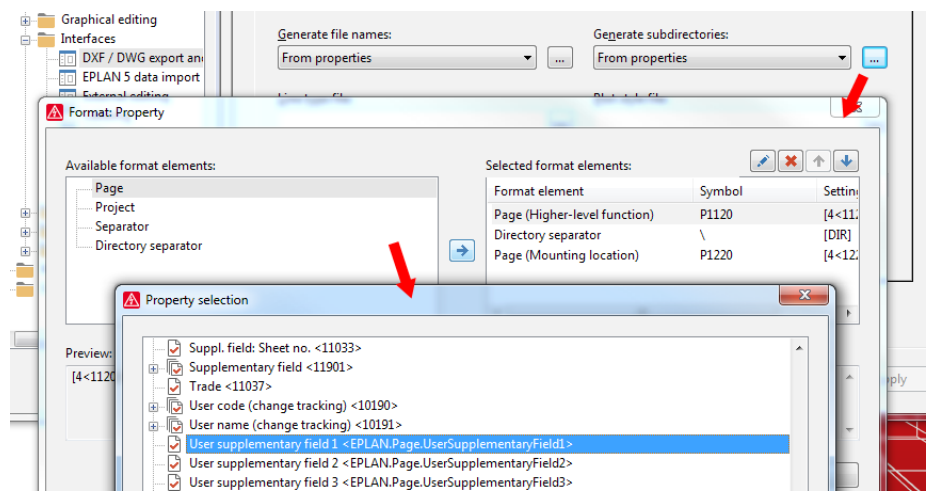


Fig. 8.10 Option of output in subdirectories

The two remaining settings, Line type file (*.lin) and Plot style file (*.ctb), are provided by the target system and must exist as files for EPLAN so that they can be included in reports as well.

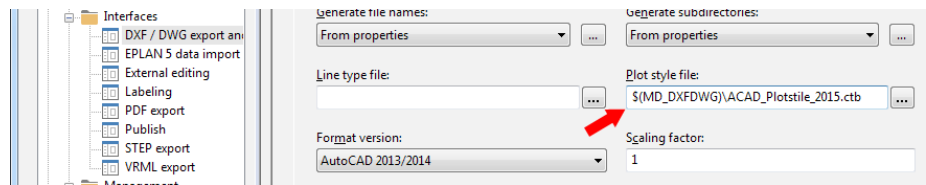


Fig. 8.11 Options to let the target system define types/styles

You exit the **Settings: DXF/DWG export and import** dialog by clicking the **OK** button. The **CANCEL** button discards all changes you may have made to the scheme settings. EPLAN applies the newly defined scheme in the **DXF/DWG export** dialog and adjusts the target (directory) for the output, if it was changed in the scheme. When you click the **OK** button, EPLAN generates a DXF or DWG file for the selected pages using the defined output parameters.

NOTE: If this should result in duplicate file names, EPLAN will display a message and then cancel the export without any further messages.

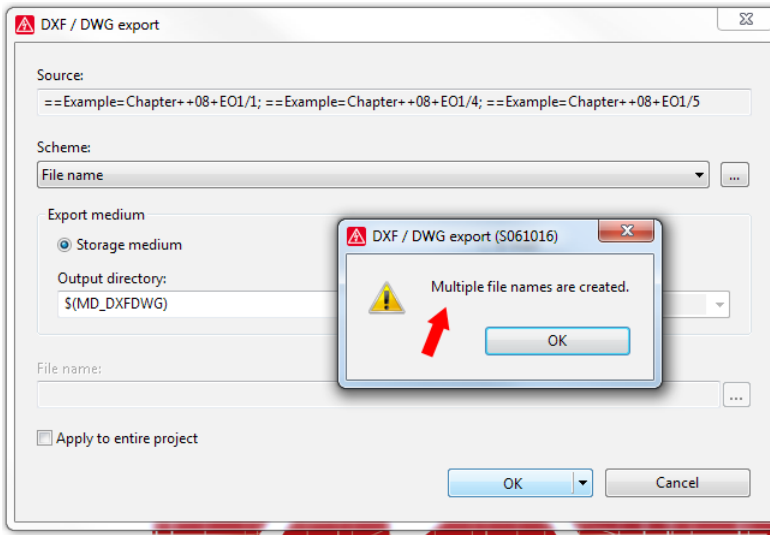


Fig. 8.12 Error message when generating the output

If the output goes through without any problems, the output will be generated. EPLAN then closes all dialogs, and again focuses on graphical editing.

DXF_DWG			
Bernd Gischel			
Handbook EPLAN Electric P8			
Chapter			
8.6.1			
Name	Änderungsdatum	Typ	
☐ 1_Export.dxf	05.06.2015 15:57	DXF-Datei	
☐ 4_Export.dxf	05.06.2015 15:56	DXF-Datei	
☐ 5_Export.dxf	05.06.2015 15:56	DXF-Datei	

Fig. 8.13 Generated DXF files in the stored directory structure

8.1.2 Import of DXF and DWG files

In addition to exporting (generating) DXF/DWG files, EPLAN can import CAD drawings in DXF and DWG formats and create project pages from them.

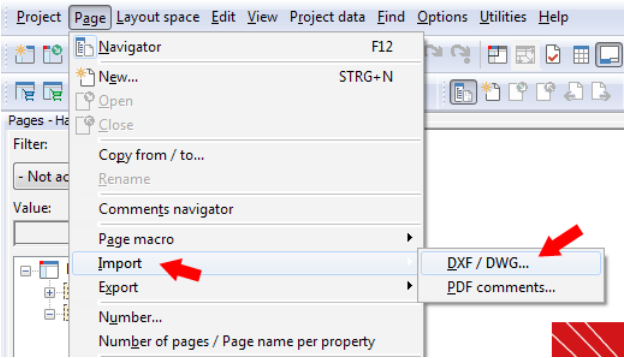


Fig. 8.14 Import DXF/DWG files

The **DXF/DWG file selection** dialog is opened via the **PAGE/IMPORT DXF/DWG** menu. By default, EPLAN opens the DXF/DWG directory defined at installation for selecting the files. You can, of course, select any other directory.

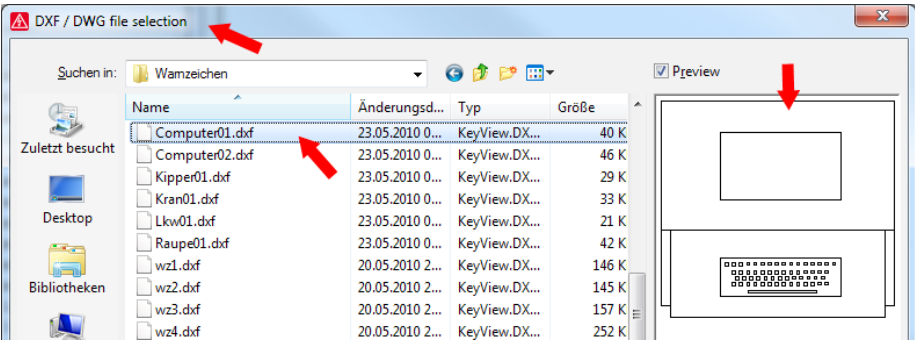


Fig. 8.15 Dialog for importing DXF or DWG files

The desired file type for the import (DXF or DWG) is set in the lower area of the dialog.

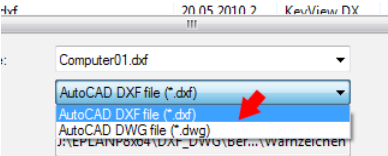


Fig. 8.16 Selection of the file type

You can switch on the **PREVIEW** option to make initial selection of the file easier. EPLAN then displays a small preview of the relevant file. If you select multiple files, then no preview is shown.

You can now select one or more files in the **DXF/DWG file selection** dialog and apply this selection by clicking the **OK** button. EPLAN opens the **DXF/DWG import** dialog.

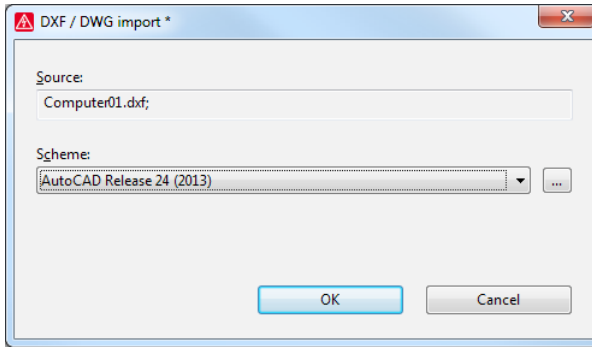


Fig. 8.17 Import dialog

The scheme set in the lower area is the same as that used to export DXF/DWG files (because this scheme also contains the import setting).

To adjust the import scheme, or create a new one, you click the [...] button to open the now-familiar **Settings: DXF/DWG export and import** dialog.

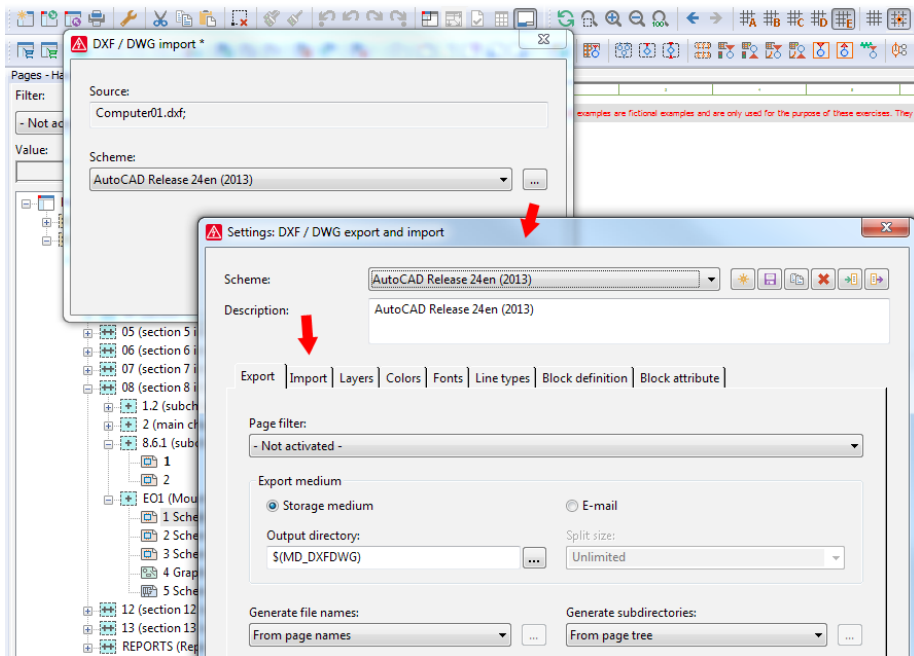


Fig. 8.18 Settings for importing DXF / DWG files

On the **Import** tab you can now define parameters and options for importing the DXF/DWG files before starting the actual import.



TIP: It is a good idea to initially accept the settings as they are defined in the EPLAN default scheme. You can specify the **Generate pages** setting in advance, if you wish. However, this is queried again in a later dialog, with more options for placing the DXF/DWG files at specific locations in the project.

If you have made no changes, you can click **CANCEL** to exit the dialog. EPLAN returns to the **DXF/DWG import** dialog. When you click **OK** here, EPLAN starts the import. This is followed by the **Assign pages** dialog, which asks for important information defining how the DXF/DWG files are to be sorted into the existing page structure of the project.

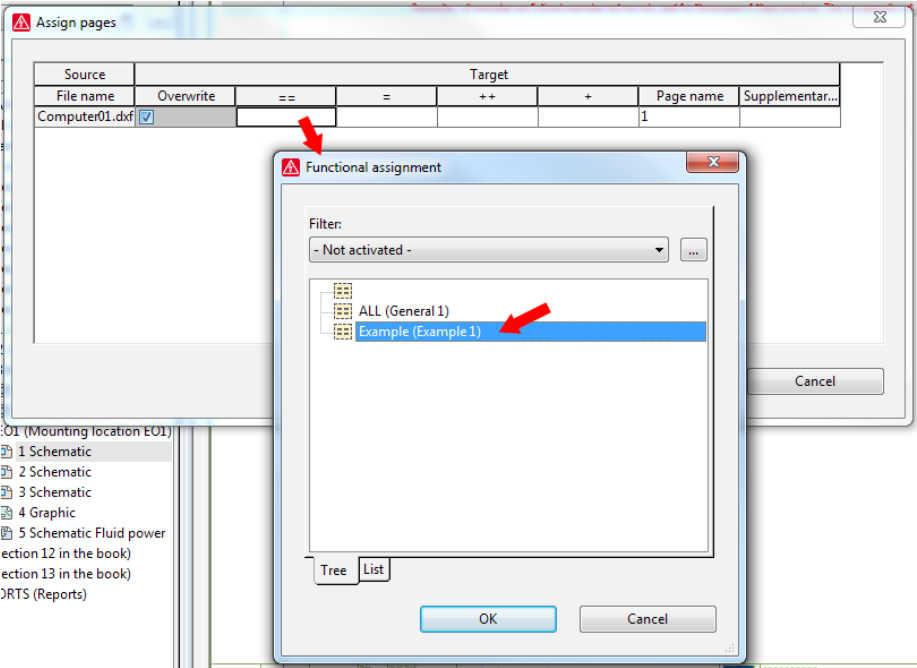


Fig. 8.19 Assign pages prior to the actual generation of the project pages

You enter the desired target designations into the dialog. You can use the **NUMBER** button to renumber the target pages before importing them if necessary.

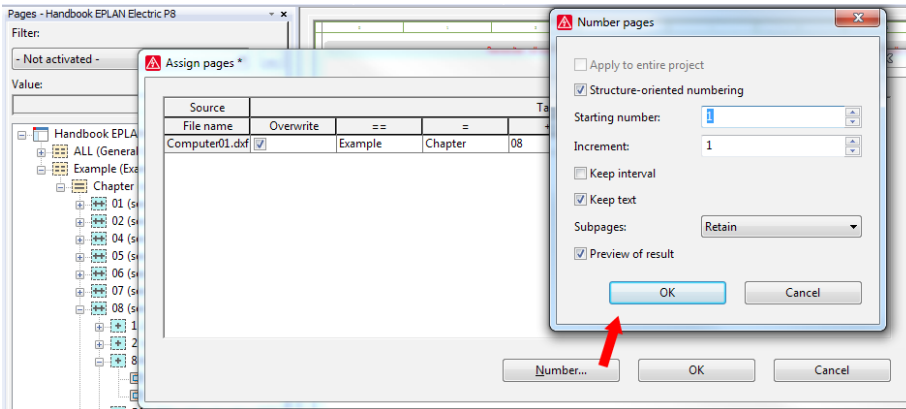


Fig. 8.20 Numbering pages automatically

When you click OK in the **Assign pages** dialog, the pages are imported (possibly accompanied by an **Import formatting** prompt, which can be confirmed by clicking OK) and sorted into the page structure of the project as specified.

This completes the import of the DXF/DWG files. The DXF/DWG files are read in and displayed on pages of the graphic type. They can now be edited in the usual manner with the normal graphical editing functions.

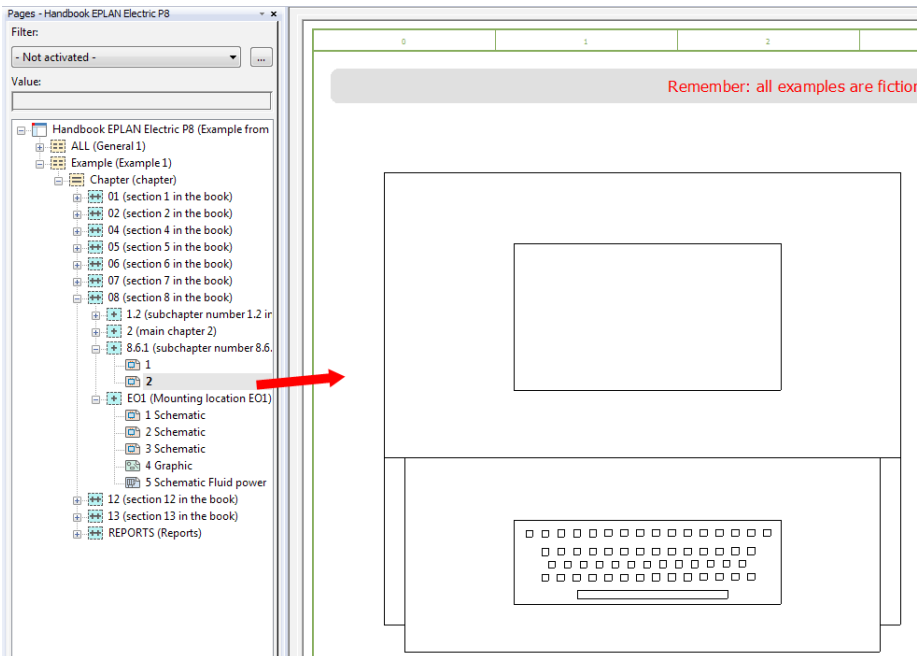


Fig. 8.21 Read-in example of a DXF file

Just to remind you one more time: These pages are graphical pages only, without any (electrical engineering) logic. When schematic pages are exported in the DXF/DWG format, the pages are “dismantled” into graphics.

If these DXF/DWG files are then imported (e.g. you import the previously exported schematics), then after importing they are still graphics files and therefore all the imported elements on the pages are also graphics. These imported DXF/DWG files are **not** converted back to logic pages.

8.2 Image files

In addition to DXF/DWG export and import for various CAD systems, EPLAN can also export project pages as image files.

Schematics and all other pages belonging to the project, such as reports, cover sheets, or the table of contents, are normally always exported as image files when they need to be archived.

The DXF/DWG formats, for example, are not suitable for long-term archiving because these formats may change over time. Image formats on the other hand will remain unchanged in the future and are recommended for long-term archiving.

8.2.1 Exporting image files

You export image files in a similar manner to exporting DXF/DWG files.

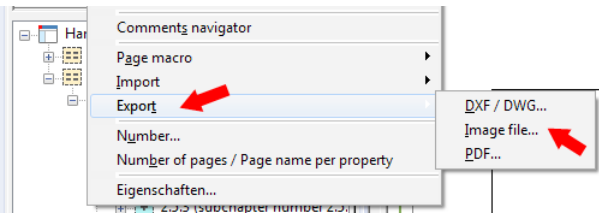


Fig. 8.22 Page/Export/Image file menu

You first select the pages to be exported in the **page navigator**. If you wish to export the entire project, then you just need to select the name of the project in the page navigator.

NOTE: The **Image file export** function also only allows selections from a single project. You cannot select pages for export from different projects at the same time.

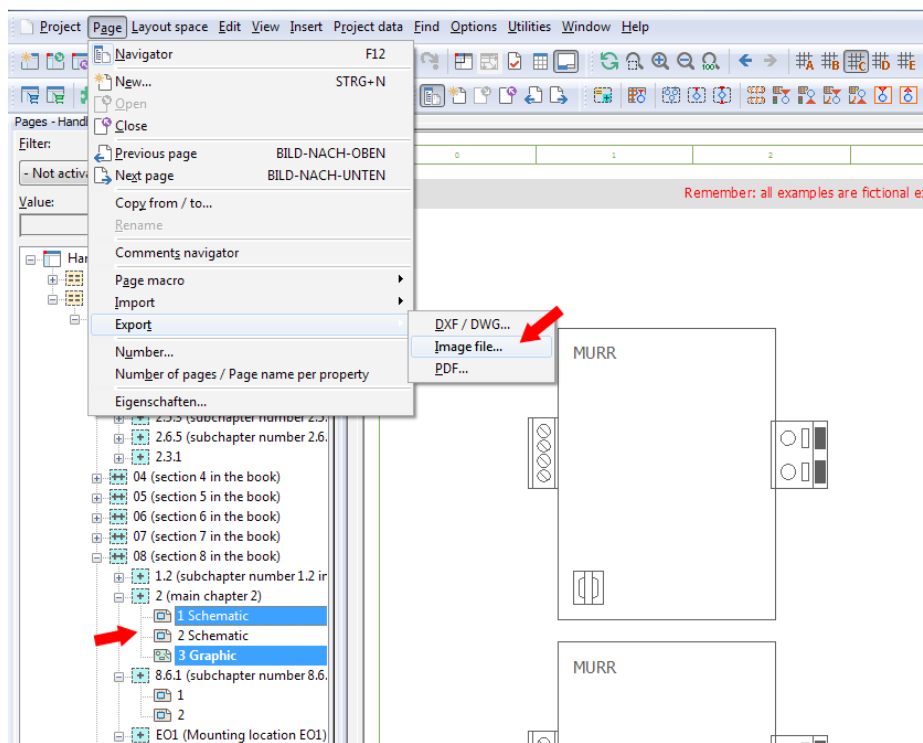


Fig. 8.23 Selected pages in the page navigator

You use the **PAGE/EXPORT/IMAGE FILE** menu to open the **Image file export** dialog.

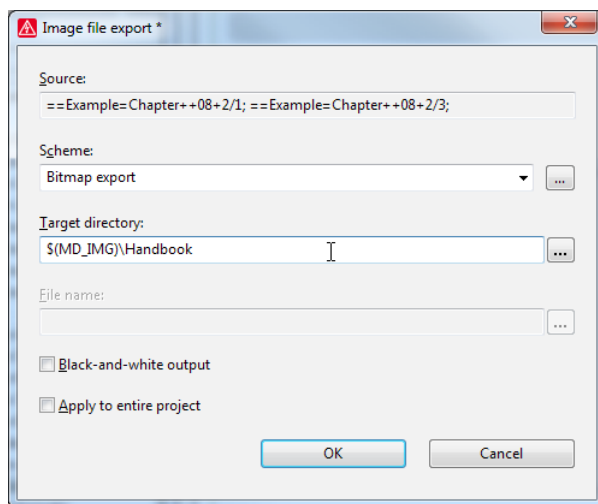


Fig. 8.24 Image file export dialog

In this dialog, you can now select and apply an existing scheme that defines the output. EPLAN also allows you to define the output settings, such as compression or the color depth of the generated image files, via your own scheme.

To create your own scheme for outputting image files, click the  button next to the **Scheme** selection field. EPLAN opens the **Image file export** settings dialog.

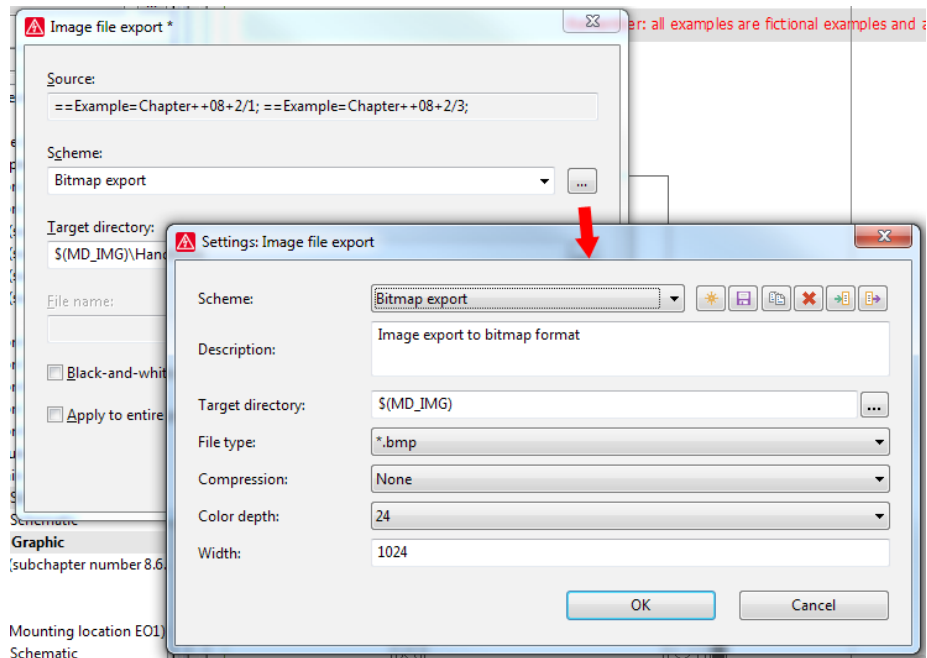


Fig. 8.25 Settings for image file export

Click the graphical **NEW** button to create a new scheme. In the subsequent **New scheme** dialog, you define a name for the new scheme and an optional description. When you confirm and apply your entries by clicking **OK**, the **Image file export** dialog is displayed again. The new scheme name is now loaded by default and you can edit it.

Make sure to click the **SAVE** button to save the scheme after you have made your settings, such as the selection of the file type, the compression, or color depth. You can now exit the dialog via the **OK** button.

EPLAN now returns to the **Image file export** dialog. In addition to allowing selection of a different scheme or output directory, this dialog has two further options.

☐ Black-and-white output

The first option affects the output of the pages themselves: Should the output be generated in color or black and white? Selecting **Black and white output** generates image files without color information.

Exceptions to this are embedded images, such as a company logo in a plot frame or cover page form. These are usually output in the original colors. If the black-and-white output option is not used, then EPLAN generates the output pages in color. In this case, “color”

means that the output in the image file looks the same as that which you see on the screen.

The second option **Apply to entire project** affects the range of the output pages. At this point you can decide to export all the project pages to image files, even if you have only selected some pages and not the whole project for exporting. You do not need to cancel the export, you can simply activate the **Apply to entire project** option.

☐ Apply to entire project



TIP: You can directly enter a new directory into the *Target directory* field (output directory). EPLAN can also create a *[New directory]* automatically on export. If EPLAN cannot find the output directory then it creates one.

If all settings have been made, EPLAN can start the export. Exit the dialog via **OK**. EPLAN now generates the individual image files and stores them in the set output directory.

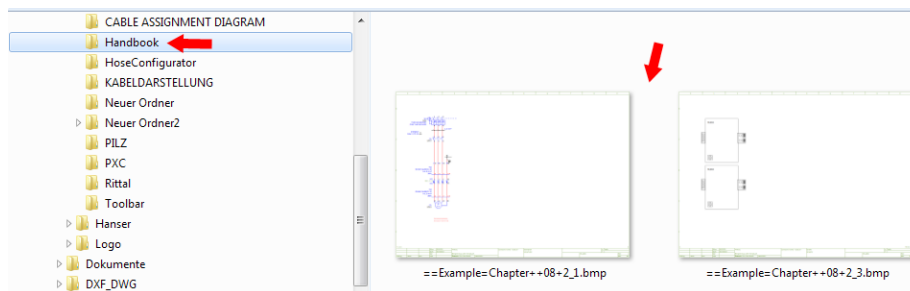


Fig. 8.26 Exported schematic pages as images

8.2.2 Insert image files (import)

You cannot import image files as pages in EPLAN. The **PAGE/IMPORT** menu has no *import image files* function. You cannot compare this type of function with the function for importing DXF/DWG files. They are completely different functions.



NOTE: EPLAN handles the import of image files in a different way. Image files are not imported; they are inserted and their size or resolution can be changed later.

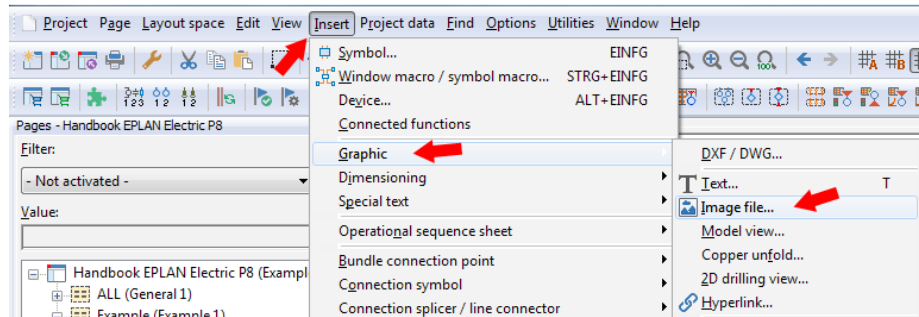


Fig. 8.27 Insert image files menu

When you select the **INSERT/GRAPHIC/IMAGE FILE** menu, EPLAN opens the **Select image file** dialog. As usual in EPLAN, the corresponding default user directory is displayed (in this case for images). Other directories can, of course, be selected.

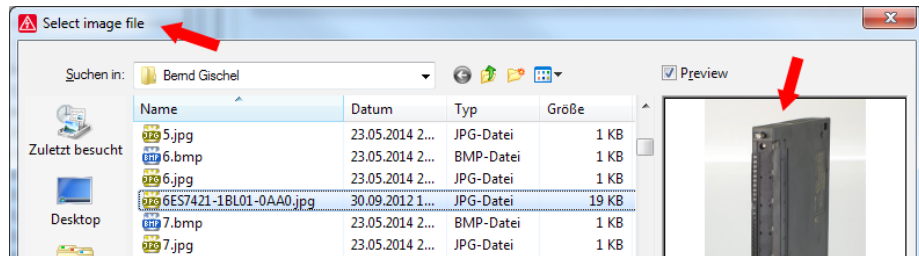


Fig. 8.28 Dialog for the selection of image files

EPLAN can insert numerous different image formats. The corresponding image format is preselected in the **Select image file** dialog in the lower area.

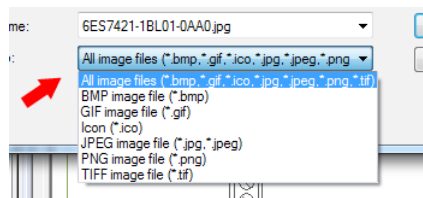


Fig. 8.29 Selection of the image format

File types such as **.bmp* and **.jpg* are the most common, but space-saving formats such as **.png* are also possible. There are no known limitations on the directory structure or file names used. Special characters permitted by the operating system or deeply branched directory structures (including spaces) are no problem for EPLAN.

After selecting the desired image file (only one image file at a time can be inserted), click the **OPEN** button. The important **Copy image file** dialog is displayed.

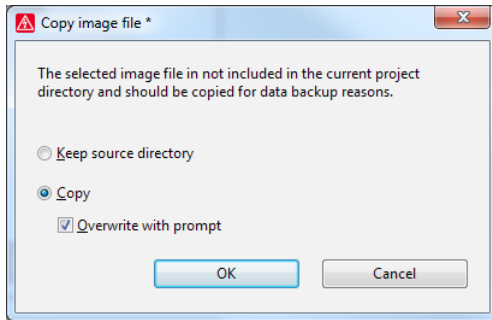


Fig. 8.30

Copy image file



TIP: You should take the message in this dialog seriously. If the image file is not copied to the project directory (the **Keep source directory** option), then the possibility exists that this “linked” (referenced) image file will not be saved during a data backup (depending on the settings). It is, therefore, a good idea to always use the **Copy** option in this dialog.

If necessary, you can enable the **Overwrite with prompt** option. If an image file with the same name already exists in the target directory, then a security message is displayed before overwriting. You can still cancel the copy operation at this point.

If you click OK in this dialog, then EPLAN copies the image file into the `\IMAGES` project directory and returns to graphical editing. You now define the first corner of the image by pressing the ENTER key, or by clicking the left mouse button.

EPLAN displays a window that you can enlarge by pulling with the mouse or via the cursor keys. The size of this window defines the initial size of the image on the page.

Insert an image into a page

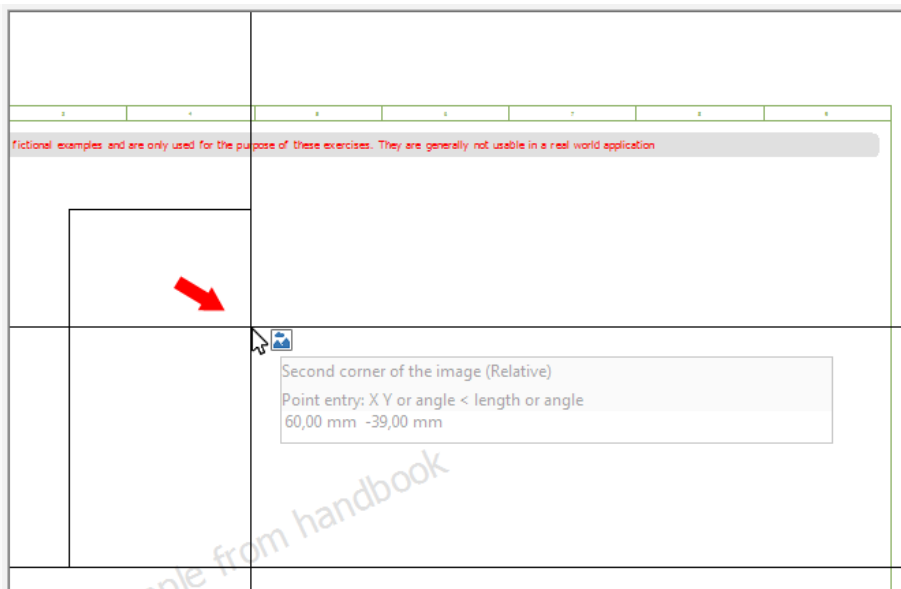


Fig. 8.31

Pulling open a window

Clicking the left mouse button or pressing **ENTER** again causes EPLAN to display the **Image properties** dialog. Here you can make settings defining the **size in coordinates** or in **percent** of the original size.

The *Keep aspect ratio* setting is important. This avoids distortion of the image. This setting should always be activated.

Properties of an image

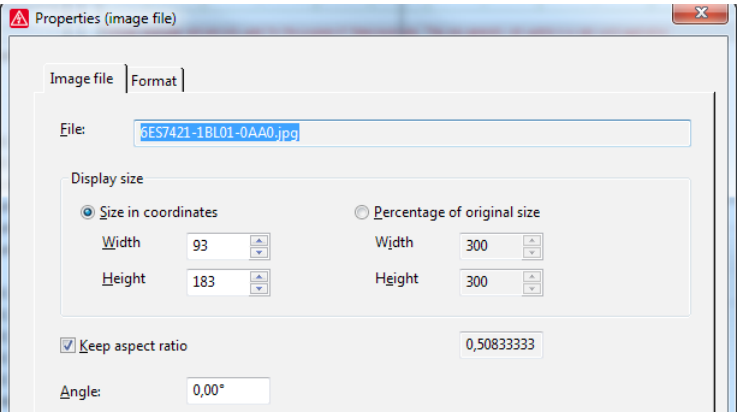


Fig. 8.32 Formatting options dialog

After you confirm the settings by clicking **OK**, the image is inserted (placed). To edit images at a later date, for example to change the size or enter something into the **Description** field, you just select the graphic with a left click. EPLAN changes the display and “draws” a box with small handles around the graphic.

Each of these handles can be used with the mouse to make the graphic larger or smaller. You can, of course, specify the size of the image directly in the image properties at any time. A new layer in EPLAN also lets you move the graphic into the background where you can no longer use edit functions such as delete, move, etc. if the *Background* option in layer management is selected. This layer, *EPLAN705.Graphic.Background*, is selected or set in the properties dialog on the **Format** tab in the **Layer** selection.

To open the **Properties** dialog, you first select the graphic as described above and then press the **CTRL + D** shortcut key.

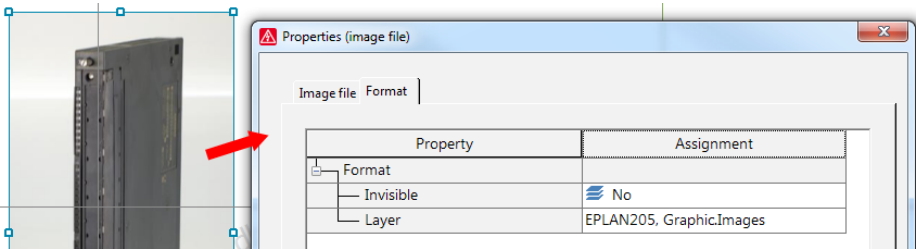


Fig. 8.33 Properties dialog

This function for calling up the properties also exists in the right-click popup menu.

8.3 Print

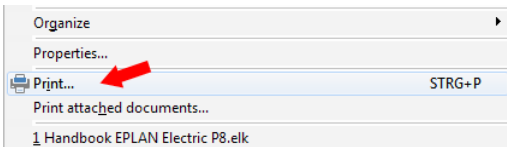


Fig. 8.34 The Print menu

The print command for pages in EPLAN is accessed via the **PROJECT/PRINT** menu item. The **Print** dialog offers a number of settings that will now be briefly described. The page print settings should be adjusted initially to suit your particular printer.

8.3.1 The Print dialog and its options

If you wish to print pages, then you first select them in the **page navigator** and then prepare them via the **PROJECT/PRINT** menu. EPLAN then displays the **Print** dialog.

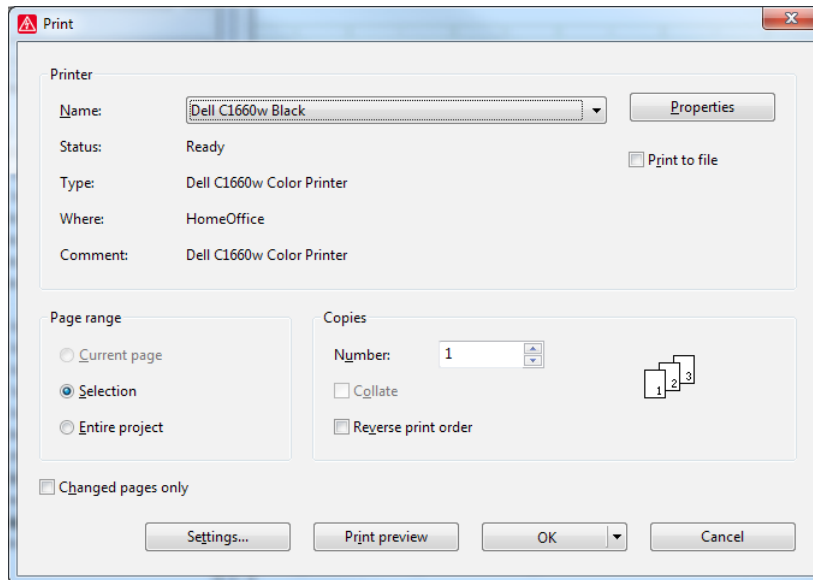


Fig. 8.35 The Print dialog

With just a few exceptions, the **Print** dialog is very similar to the conventional, familiar Windows print dialogs. Familiar options are the selection of the installed printer and the option allowing printing of multiple copies with different output sorting options.

8.3.1.1 Page range/Changed pages options

At this point, in the **Page range** options, you can select the **Selection** option to print only the currently selected pages in the page navigator or change your mind and select the **Entire project** option. The **Current page** option is automatically set to deactivated as soon as more than one page has been selected in the page navigator. The **Changed pages only** setting prints only those pages that have changed since the last print operation.

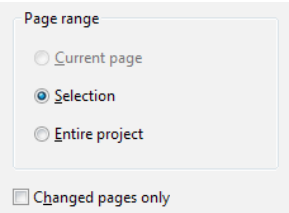


Fig. 8.36
Print options

8.3.1.2 The Settings button

The **Settings** button contains a number of print settings for the pages themselves. Here you can define the print size, whereby **Print to scale** means: If a page was assigned the A3 plot frame, for example, then it is printed 1 : 1 in A3 format. Printers that are smaller than A3 format (i.e. generally A4 printers) will only print the upper left corner of such pages.

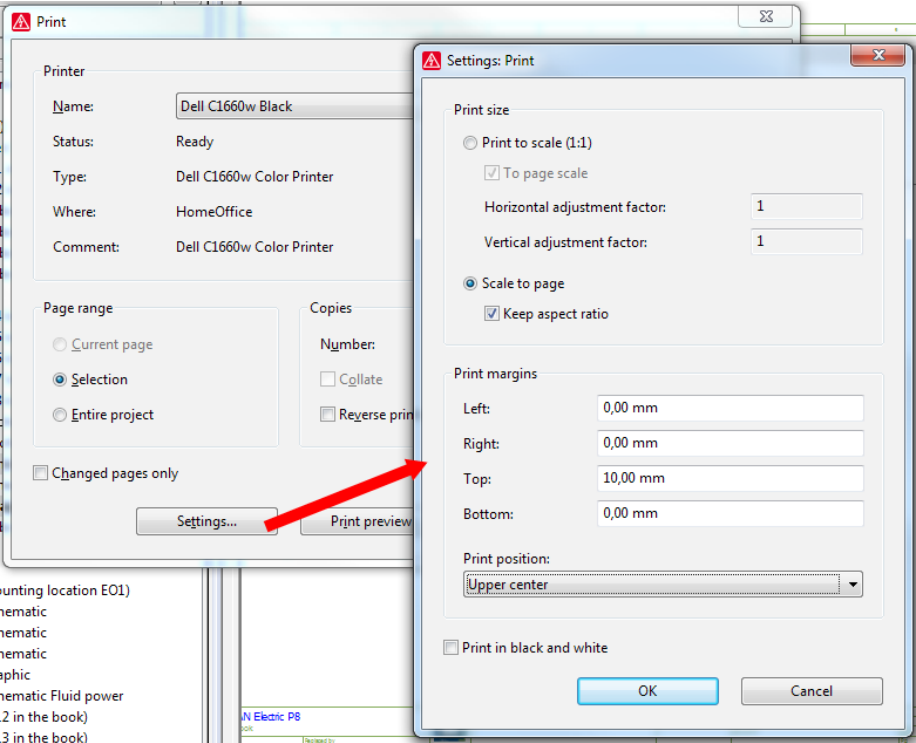


Fig. 8.37 More page settings

The **Scale to page** setting and the **Keep aspect ratio** parameter should be used by default. This causes all larger dimensions to be scaled down to the printer model and a complete page is always printed. Scaling usually is to an A4 page size.



NOTE: The **Print margins** settings should initially be adjusted to suit the installed printer(s).

Unfortunately, one cannot provide generally applicable suggestions for the print margins because there are simply too many printer models with widely varying technical properties. Here you can only try things out and make a number of tests with the connected printer.

Selecting the **Print position** allows you to position the page on the printed sheet of paper. EPLAN offers a number of different print positions. It is a good idea to use the **Upper center** setting with an upper print margin of 10 mm.

The **adjustment factors** offer an additional setting to compensate for small deviations in printouts to scale. Here, too, it is a good idea to experiment a bit, because these values depend on the printer model as well as on the printer driver in use. It is therefore not possible to specify generally applicable values here.

8.3.1.3 The Print preview button

The **Print preview** button in the **Print** dialog provides a rough preview before performing the actual printing.

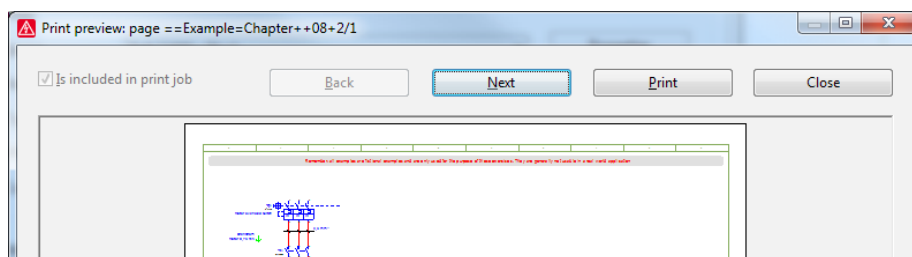


Fig. 8.38 Print preview

However, not all the information on a page is displayed. This is not so important because, at this point, the print preview is mainly used to check the position of the page. If the print preview looks OK, you can print directly from print preview via the **PRINT** button. Without further prompts, EPLAN will start printing on the printer specified as the default printer in the operating system

8.3.2 Important export/print setting

Aside from the simple print/export and the print settings in the **Print** dialog, there are some other important user settings related to printing and/or exporting the project. As the name implies, these settings are always user-defined.

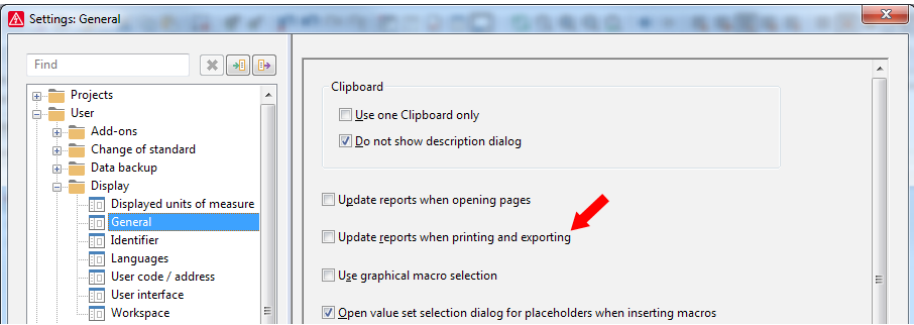


Fig. 8.39 Update reports when printing/exporting settings

To ensure that reports are current, the **Update reports when printing and exporting** setting should be activated. This prevents the project from being printed with outdated reports. This setting is located in the **OPTIONS/SETTINGS/USER/DISPLAY/GENERAL** menu.

8.4 Export and import of projects

In addition to exporting and importing image files or DXF/DWG files, EPLAN also allows entire projects to be imported and exported in XML format.

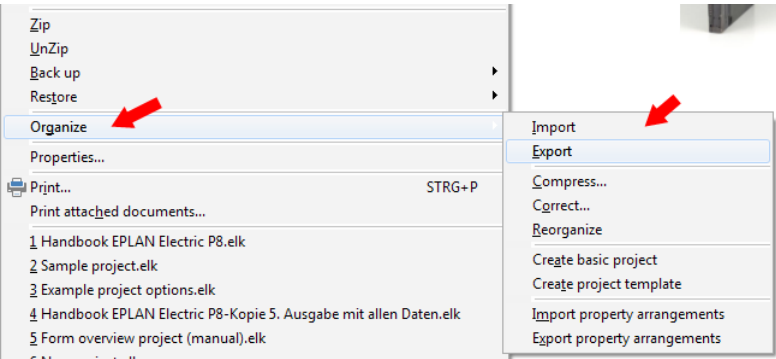


Fig. 8.40 Import and export project data

8.4.1 Exporting projects

Projects are exported from the **page navigator** via the **PROJECT/ORGANIZE/EXPORT** menu.

To export a project, you first select the desired project and then start the **EXPORT** function. When the *Export* function starts, the **Browse For Folder** dialog is displayed. This is where you define the directory where you want the exported project to be saved.

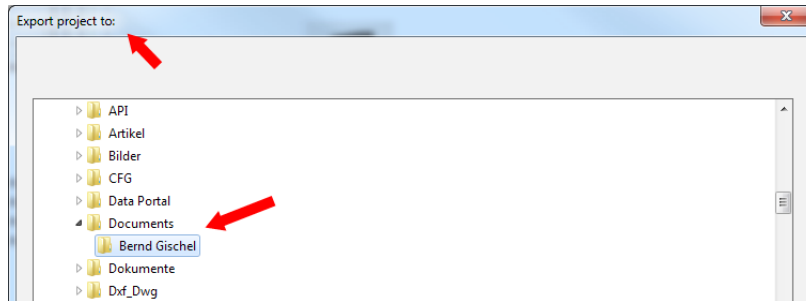


Fig. 8.41 Defining the export folder



NOTE: Using the page navigator, it is only possible to export **one** project at a time. If several projects are selected, then the *Export* function cannot be chosen (the entry is grayed out).

After you selected the export folder and click **OK**, EPLAN immediately starts the export without further prompts. EPLAN maps the progress in the *Export project* dialog. The project is exported and is assigned the file extension **.epj*.

8.4.2 Importing projects

The counterpart to exporting is importing XML projects. In the graphical editor, you execute this function via the **PAGE/ORGANIZE/IMPORT** menu.

When you select the **IMPORT** function, EPLAN opens the **Import project** dialog. Here you select from the directory structure the directory containing the project (**.epj*) to be imported.

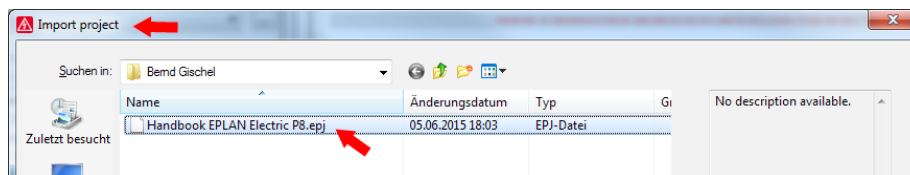


Fig. 8.42 Importing a project (*.epj)

The project is selected and applied via the **OPEN** button in the lower area of the dialog. EPLAN opens the **XML project** dialog and inserts the selected project into the **XML file** field.

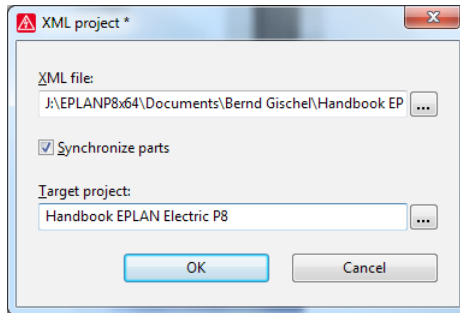


Fig. 8.43 Importing the EPL(XML) project

The **Synchronize parts** parameter enables EPLAN to synchronize project parts with the parts master data during the import. You enter the name of the project into the **Target project** field and the imported project will be shown in project management under this name.

You can also use the  button in this dialog to select a different target directory for the project to be imported. After you click **OK**, the import process starts, and EPLAN generates the project in the defined target directory.

■ 8.5 Print attached documents

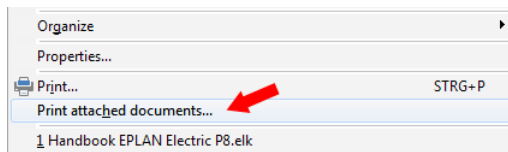


Fig. 8.44 Print attached documents

The **PROJECT** menu has a **PRINT ATTACHED DOCUMENTS** function that allows you to print all documents stored in the project, e.g. PDF documents, in a single step rather than individually.

When you select this function, EPLAN opens the **Print attached documents** dialog for the current project. In this dialog, you now have the option of excluding individual documents from printing. To do this, you remove the checkmark in the **Print** column.

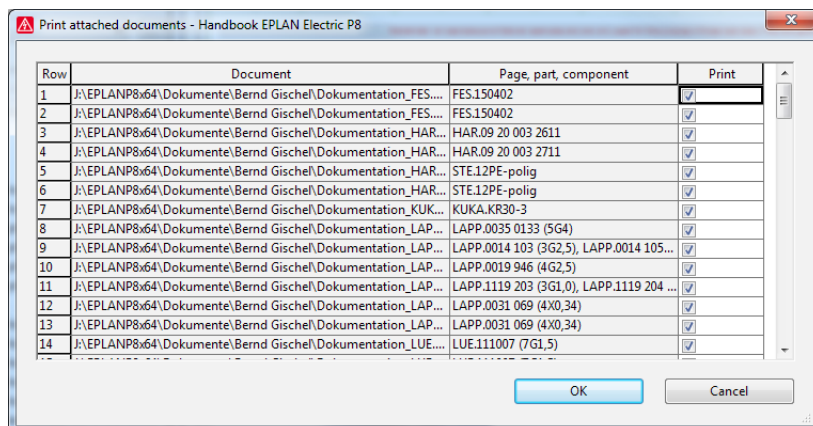


Fig. 8.45 Dialog for selection of attached documents

When you click the **OK** button, EPLAN sends all documents selected in the **Print attached documents** dialog, one after another, to the connected printer with a prompt for the printer to be selected.



NOTE: External documents (*External document* page type), linked documents on report pages and linked documents on Internet pages cannot and/or will not be printed.

8.6 Import PDF comments

EPLAN can import comments (notes, etc.) stored in PDF documents (schematics generated from EPLAN) and store these in an extra layer. This allows, for example, a service technician on-site to electronically directly add his/her own comments to the PDF schematics. These can then be imported into EPLAN, viewed, edited and evaluated.



NOTE: To be able to again import a PDF with comments into EPLAN, you will need to use EPLAN's internal PDF output function. You can find it in the **PAGE/EXPORT/PDF** menu. Write protection (read only) must be turned off for the PDF output to be generated. This setting is in the **OPTIONS/SETTINGS/USER/DISPLAY/IDENTIFIER** menu. The *Read only* setting is located on the *General* tab.

8.6.1 Importing commented PDF documents

To add comments to a PDF, the PDF generated by EPLAN must be enabled for commenting. You can do that with Adobe Acrobat Professional (version >= 7) or the freeware PDF XChange Viewer (tested with version 2.0 Build 42.4).

The comment function for Adobe Reader (version 7 and higher) can be enabled in Acrobat Professional using the **ENABLE FOR COMMENTING IN ADOBE READER** menu item in the **COMMENTS** menu. With the freeware PDF XChange Viewer, all you need to do is open the PDF, enter the comments and then save the file.

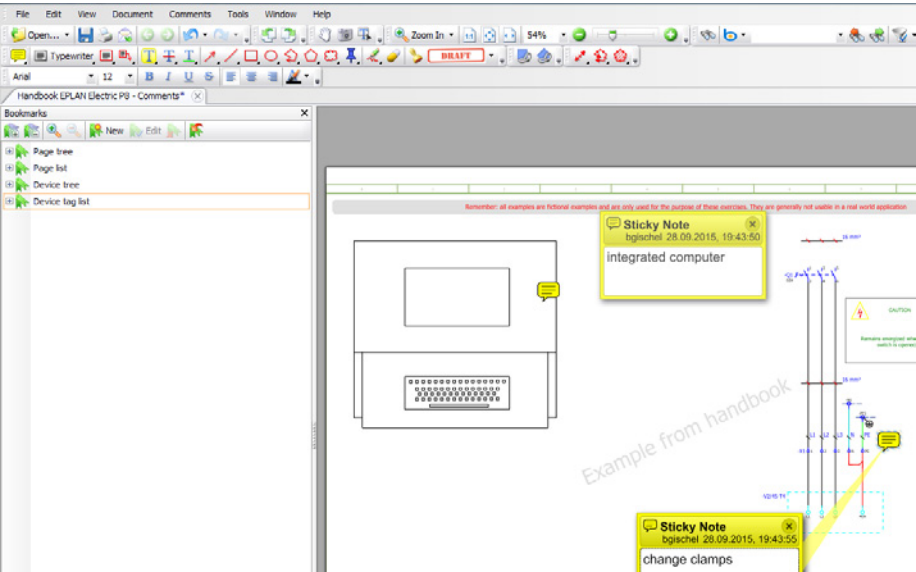


Fig. 8.46
Example of comment entries on a schematic page

After commenting the PDF document in Acrobat Reader or PDF XChange Viewer, it must be saved and closed. In EPLAN, you can now import (read in) this commented PDF document from the specified directory via the **PAGE/IMPORT/PDF COMMENTS** menu.

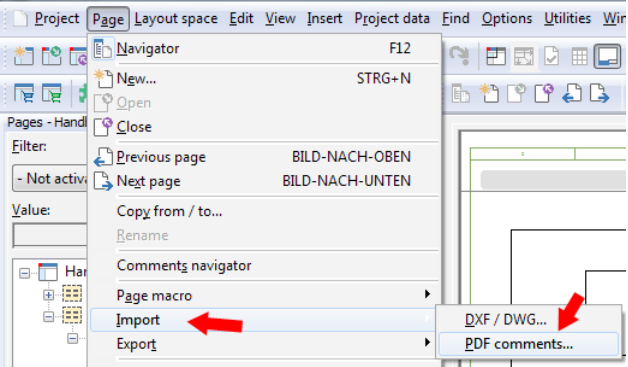


Fig. 8.47
Selecting PDF file to be imported



NOTE: To import a project, it must be in the same directory to which it was exported.

After the PDF document is imported, the **Import PDF comments** dialog containing import information is displayed.

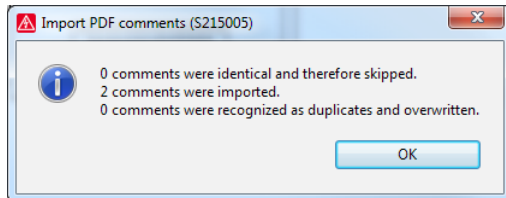


Fig. 8.48 Successful import

You can see the results of the import in the comments navigator. You can access the comments navigator from the **PAGE/COMMENTS NAVIGATOR** menu.

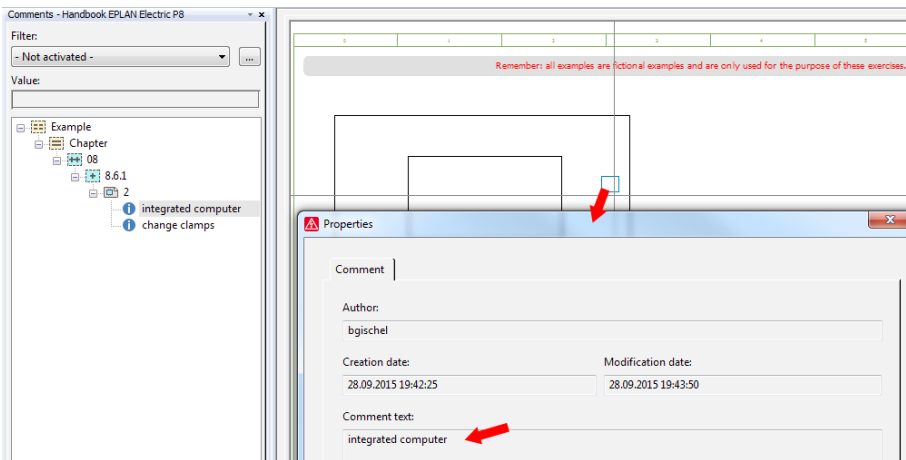


Fig. 8.49 The comments navigator

In the comment navigator, you can edit comments (e.g. set the status to “Rejected”, “Accepted”, etc.) in the comment properties.

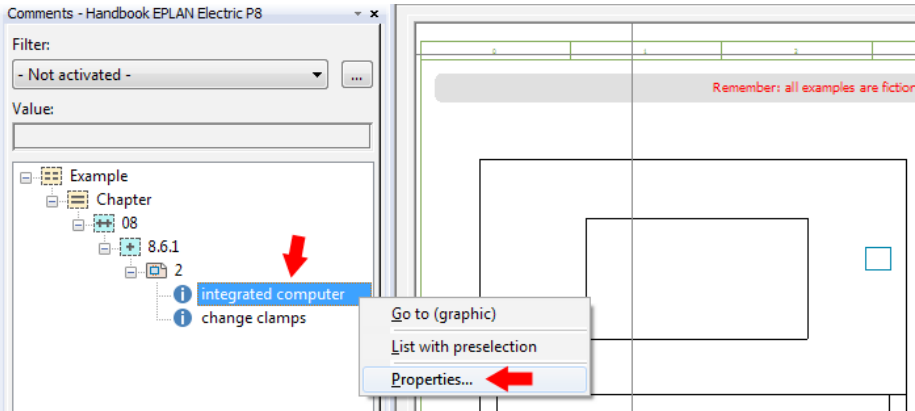


Fig. 8.50 Properties of a comment

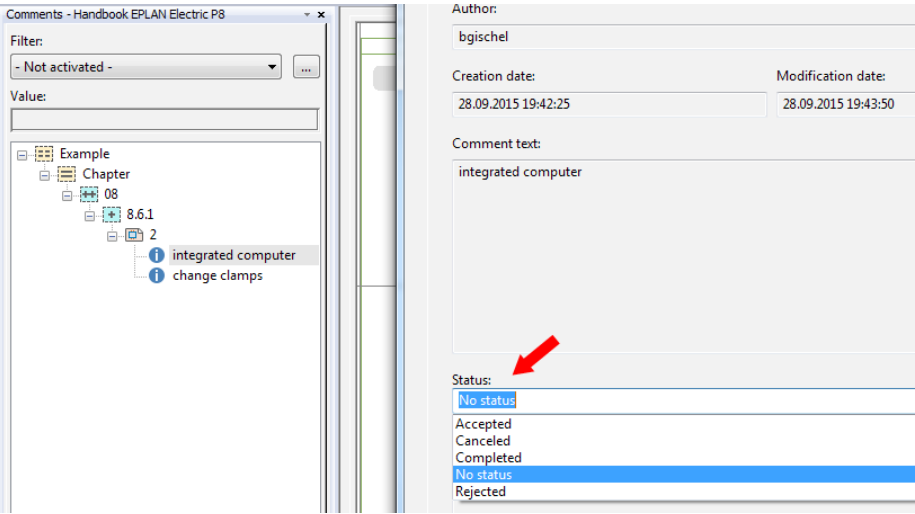


Fig. 8.51 Status of a comment

These properties are, of course, also available directly at the comments in the schematic. You can open them by double-clicking the yellow box.

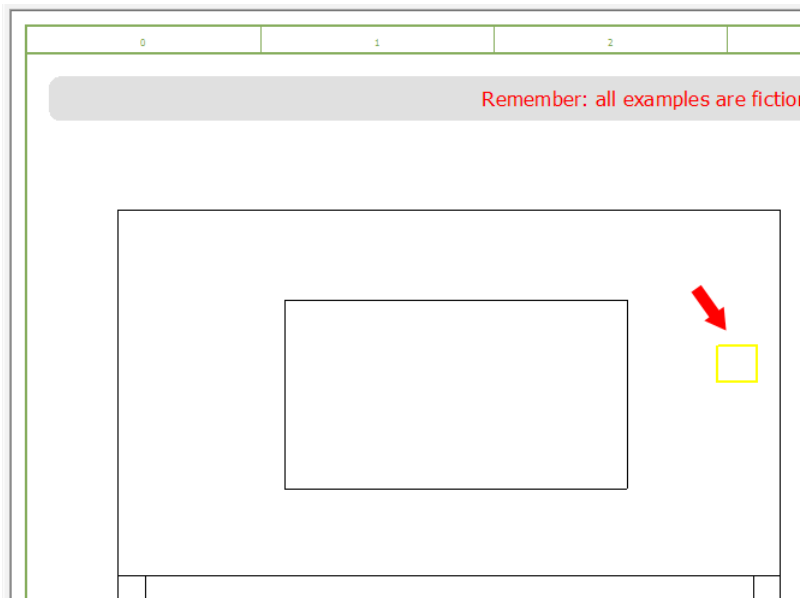


Fig. 8.52 Comment selection

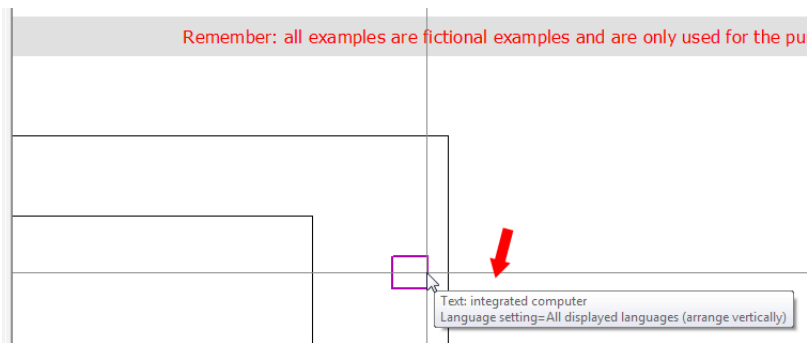


Fig. 8.53 Comment

8.6.2 Deleting PDF comments

Not only can you edit PDF comments (status of comments, etc.), you can also delete them in the usual way by selecting them and then calling the normal EPLAN delete function.

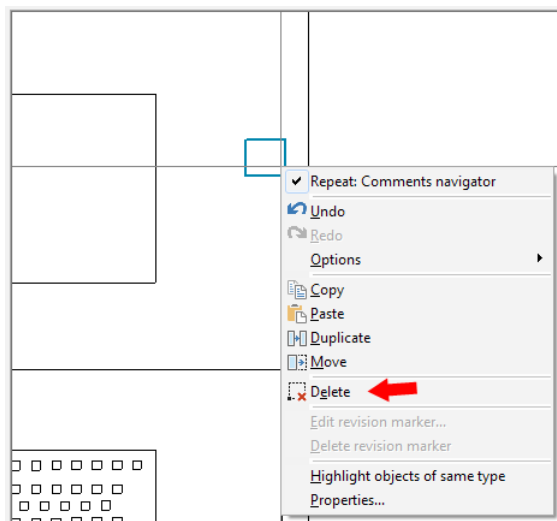


Fig. 8.54 Deleting comments

Another much more convenient method is to compress the project and create your own “Remove all PDF comments” scheme (via the PROJECT/ORGANIZE/COMPRESS menu and then modify an existing scheme or create a new one). This automatically and permanently removes all imported and unnecessary PDF comments from the project in a single step.

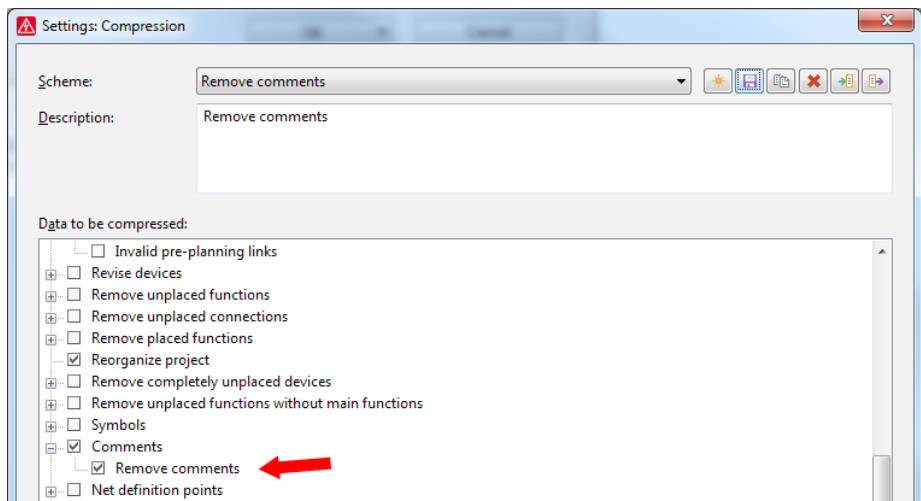


Fig. 8.55 Compression scheme to remove PDF comments

■ 8.7 Generate PDF documents

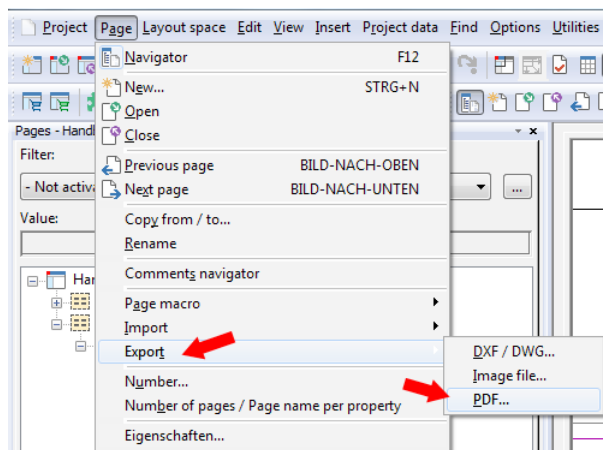


Fig. 8.56 Export as a PDF file menu

To exchange documents without problems, the PDF format is usually used. The PDF format has become a “de-facto standard” for worldwide document exchange. It therefore makes sense to also use the PDF format for exchanging schematics and other schematic documents.

The PDF format integrated into EPLAN offers more than just the simple generation of PDF documents. A certain amount of “intelligence” can be included in the documents. In EPLAN, “intelligence” means that navigation elements, such as jump functions at cross-referenced components, etc., can also be included in the PDF. Besides the intelligent PDF, EPLAN can also generate a PDF in the PDF/A format for longer-term archiving.



NOTE: If the PDF/A format is used, linked documents and models will **not** be included in the PDF output.

The setting whether the PDF is a standard PDF or one in the PDF/A format is defined in the settings for the PDF output. First select the **PAGE/EXPORT/PDF** menu and then open the scheme for the PDF output (click *More*) in the **PDF export** dialog.

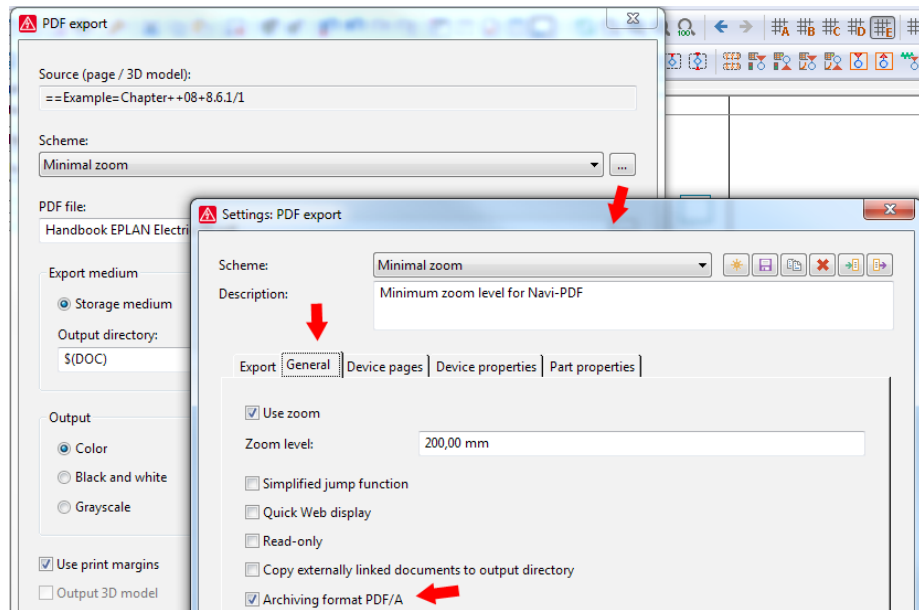


Fig. 8.57 Activate archiving format PDF/A

In the now-open scheme with the settings for the *PDF export*, open the *General* tab and activate the *Archiving format PDF/A* option.

8.7.1 Export of PDF files

Generating PDF documents is child's play in EPLAN. In the same way as when exporting DXF or image files, you first select the desired pages or the entire project in the page navigator. The PDF export is also subject to the restriction that you can only export pages from one project at a time. You cannot export pages from different projects that are open in the page navigator at the same time.

After selecting the pages, you use the **PAGE/EXPORT/PDF** menu to open the **PDF export** dialog.

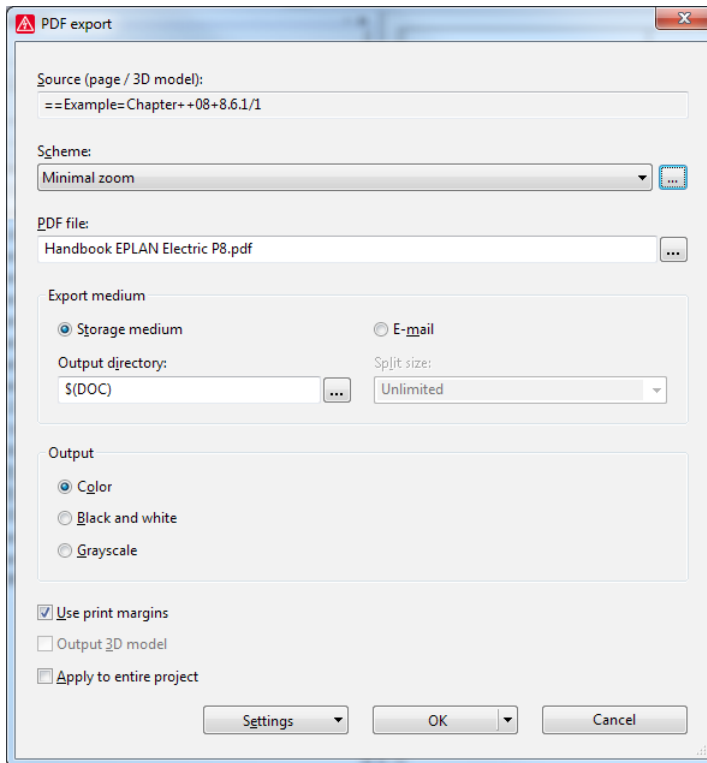


Fig. 8.58 PDF export dialog and its settings options

The **PDF export** dialog is similar to the DXF/DWG export dialog or the image file export dialog. EPLAN automatically enters the selected pages into the *Source (page/ 3D model)* field. If the project name was selected in the page navigator, then the Source field contains the project name, and the *Apply to entire project* option is grayed out.

Enter the name of the PDF file into the *PDF file* field. By default, EPLAN sets the target directory for the file name automatically to the project directory `\[PROJECT NAME]\ DOC` in the *Export medium* selection and, here, in the *Output directory* field. The PDF file to be generated also automatically receives the project name.

The target directory and the name of the PDF file can, of course, be freely chosen. Click the  button next to the PDF file field. EPLAN opens the **PDF export** dialog. Here you can use the familiar Windows functions to change the directory, create new directories, or change the PDF file name.

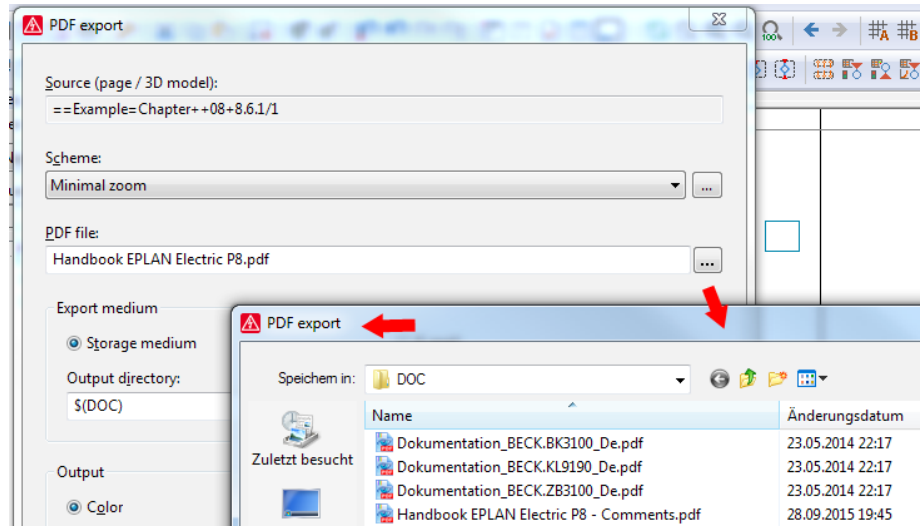


Fig. 8.59 Selection of the directory and file name for export

These modifications are saved with the **SAVE** button. EPLAN then applies the new settings of the PDF file name to the *PDF file* field.

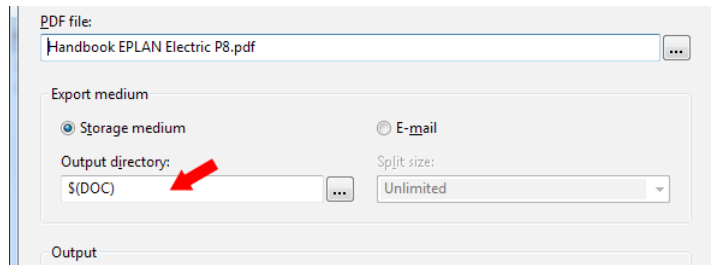


Fig. 8.60 Applied setting

In the settings of the **PDF export** dialog, it is also possible to use the *E-mail* option, including its settings (split size), directly as an export medium.

If this setting is active, EPLAN will generate the PDF, open the e-mail client and attach the PDF (zipped) to a new mail.

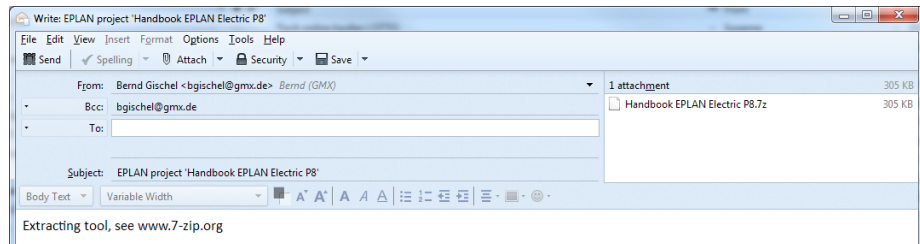


Fig. 8.61 Using the E-mail option

The PDF file can now be generated. But EPLAN also provides other settings defining the format of the output: the settings *Color*, *Black and white*, *Grayscale* as well as *Use print margins*, *Output model* and *Apply to entire project*.

You can subsequently activate the **Apply to entire project** option if you have decided to output the entire project and not just the pages you selected. If the entire project has already been selected, this setting will appear grayed out.

The **Color** output option generates a color PDF – as shown on the screen. Generally this looks quite nice, but it is unsuitable for printing on a black-and-white printer. The color settings of the background scheme *White* are applied. This setting can be found under the **SETTINGS/USER/GRAPHICAL EDITING/2D** options.

When the **Black and white output** option is activated, a black and white PDF document is output. This means that all colored elements are printed in black on a white background. However, embedded image files are output in color.

The last output option, **Grayscale**, generates a grayscale PDF. You should test this option to see what best suits your needs. I recommend this option.

The **Use print margins** setting uses the print margins defined in the workstation *print* settings. These are located under **OPTIONS/SETTINGS/WORKSTATION/GRAPHICAL EDITING/PRINT**. They can, however, be changed directly in the **PDF export** dialog at any time. To change these settings, click the **SETTINGS** button and enter your desired settings in the **PRINT MARGINS** menu item.

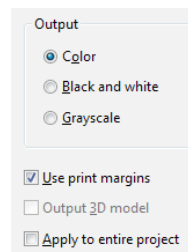


Fig. 8.62
Other settings

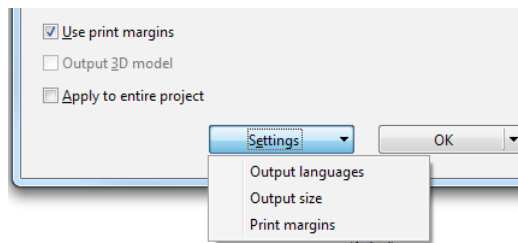


Fig. 8.63
Settings: Print margins

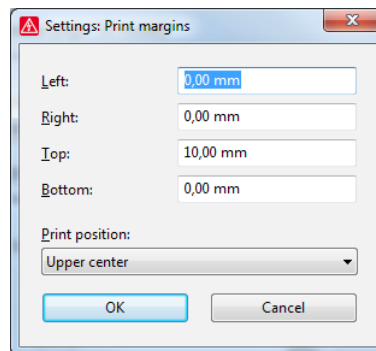


Fig. 8.64
Print margins dialog

The **Output size** setting lets you scale pages or use the original page size (the page size is determined based on the dimensions of the set plot frame) when outputting to a PDF.

The **Output model** option enables EPLAN to output the models in the project (3D panel view) at the same time.

When you have finished entering the settings, you can generate the PDF document by clicking OK. EPLAN generates the PDF document and saves it in the specified target directory. After the successful PDF export, there will be no other messages.

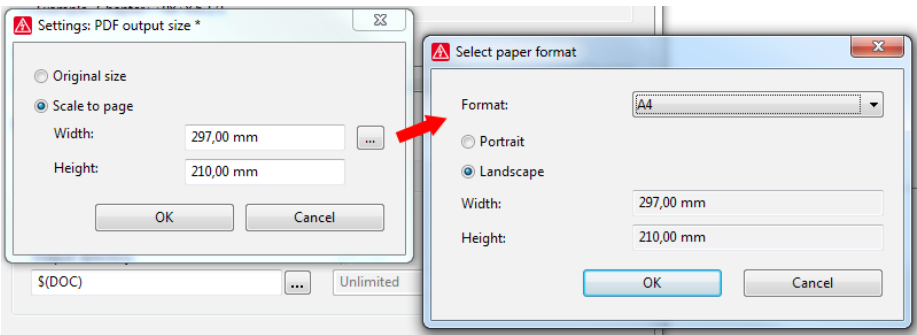


Fig. 8.65 Settings for PDF output size

8.7.1.1 Other settings for the PDF export

Aside from the “simple” settings and options for the PDF export described, EPLAN offers a great number of additional settings on the basis of a PDF export scheme, which can be used to generate information and define how the PDF document is to be generated. To open the scheme, click the *More* button in the **PDF export** dialog.

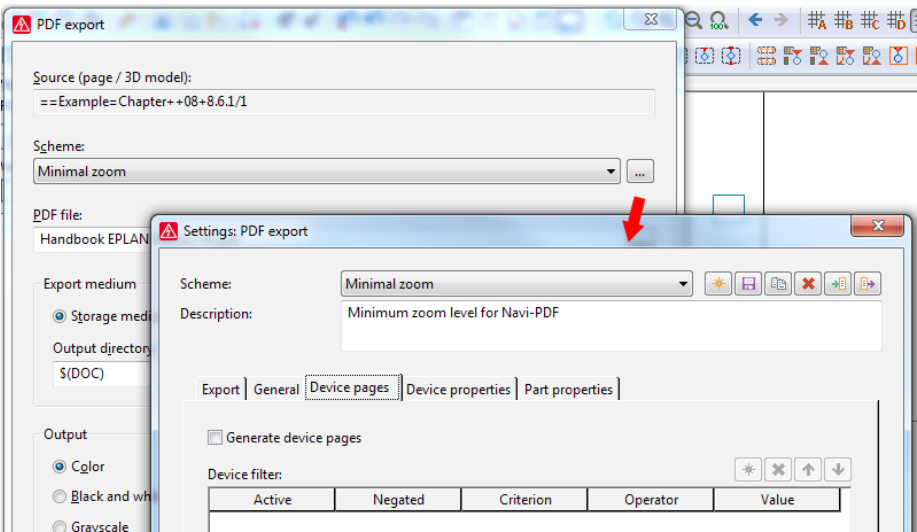


Fig. 8.66 Scheme for other detailed settings

This scheme has several tabs. The *Export* tab contains various settings about how and where the PDF is to be generated, as already described previously.

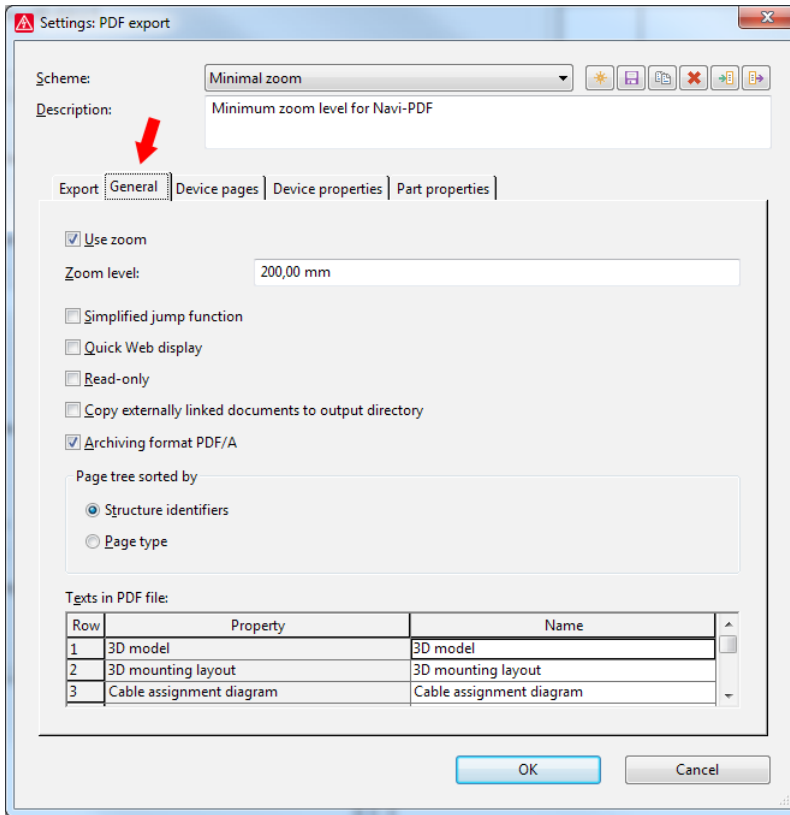


Fig. 8.67
General settings

The *General* tab contains settings and options regarding the basic functions to be given to the PDF. This includes, for example, *simplified jump functions*, *quick web display* and whether the PDF is to be sorted according to structure identifiers or page types.

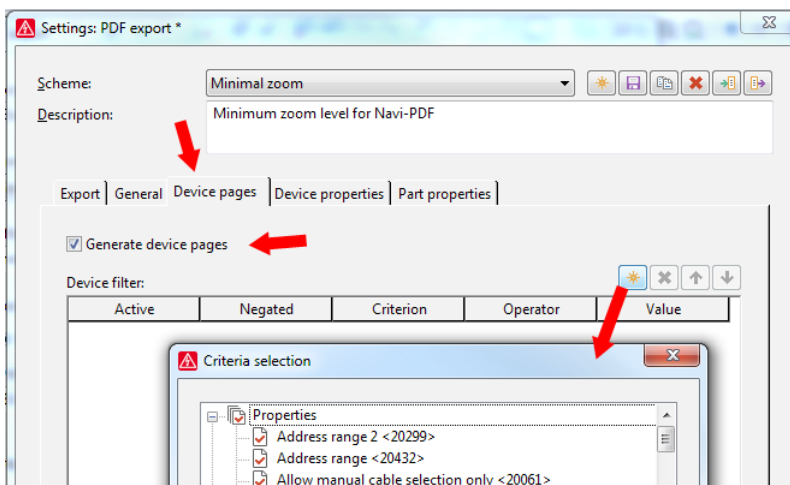


Fig. 8.68
Settings for device
pages

The *Device pages* tab defines whether device pages are to be generated at all. If so, you can make other settings to refine the level of detail. For example: Is the output also to include *device properties* or *part properties*? To which *function ranges* should these settings not apply?

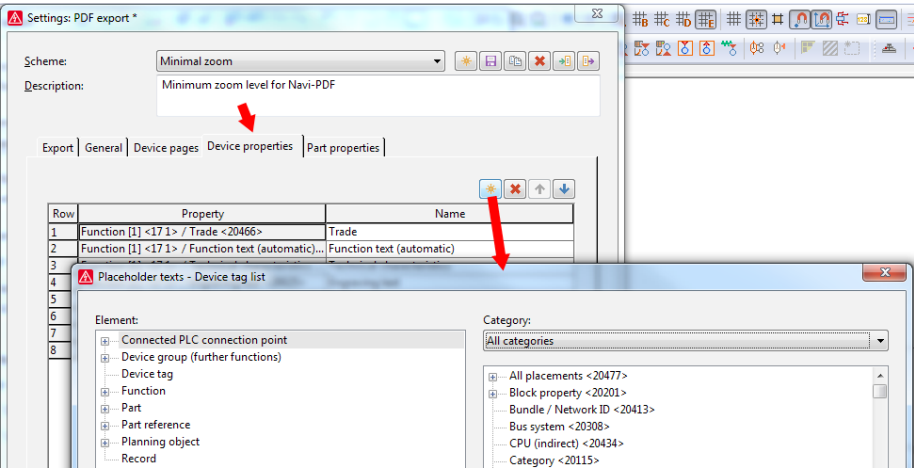


Fig. 8.69 Define device properties

Using the *Device properties* tab, you can output in the PDF very specific properties of the devices. Using the usual EPLAN approach, you can add *placeholders* here or delete them from the list.

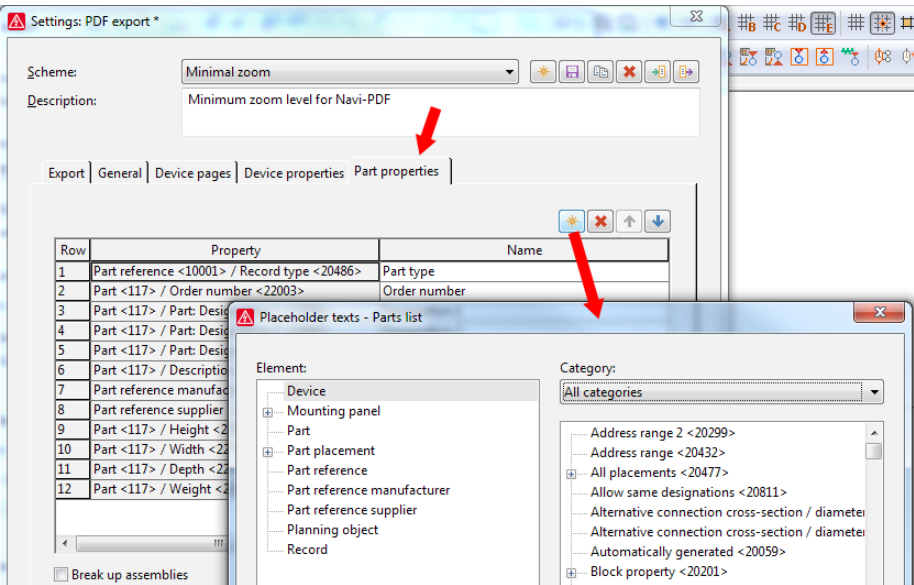


Fig. 8.70 Define part properties for the PDF export

The *Part properties* tab allows you to select the desired part properties to be written into the PDF. This could include, for example, only the *part number* and the first *designation* of a part.

If all settings have been changed accordingly and the scheme has been saved, this scheme will be available to all other projects. The settings contained in the scheme could then be used to let EPLAN generate the PDF.

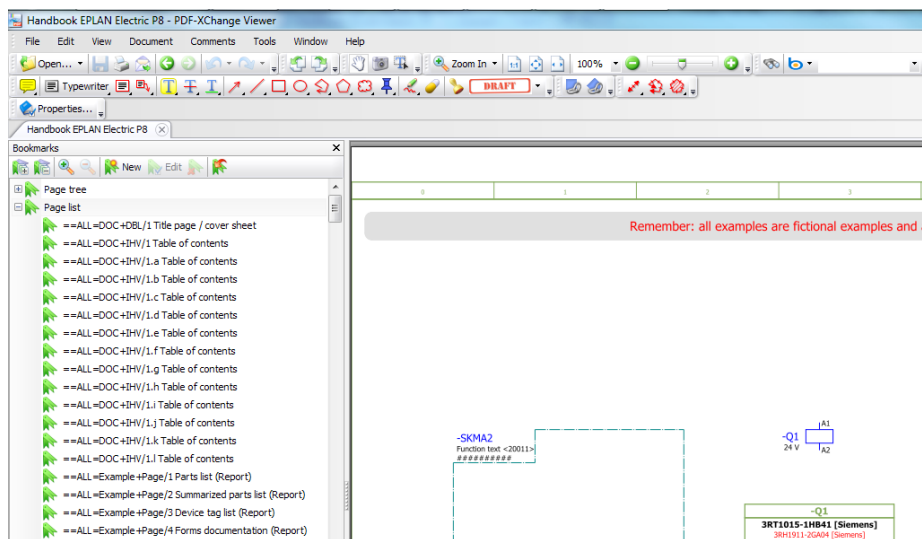


Fig. 8.71 Example: generated PDF

9

Data backup

An important feature in EPLAN is the data backup. Data backup not only includes zip-ping, unzipping, backing up, and restoring projects, there is also an e-mail function that provides a convenient way of electronically exchanging EPLAN projects between users.

Zippping, backing up, and restoring projects is not all that EPLAN is capable of, however. Individual files, such as a plot frame, a form, the parts database or dictionary (foreign language database), can also be separately backed up and restored.

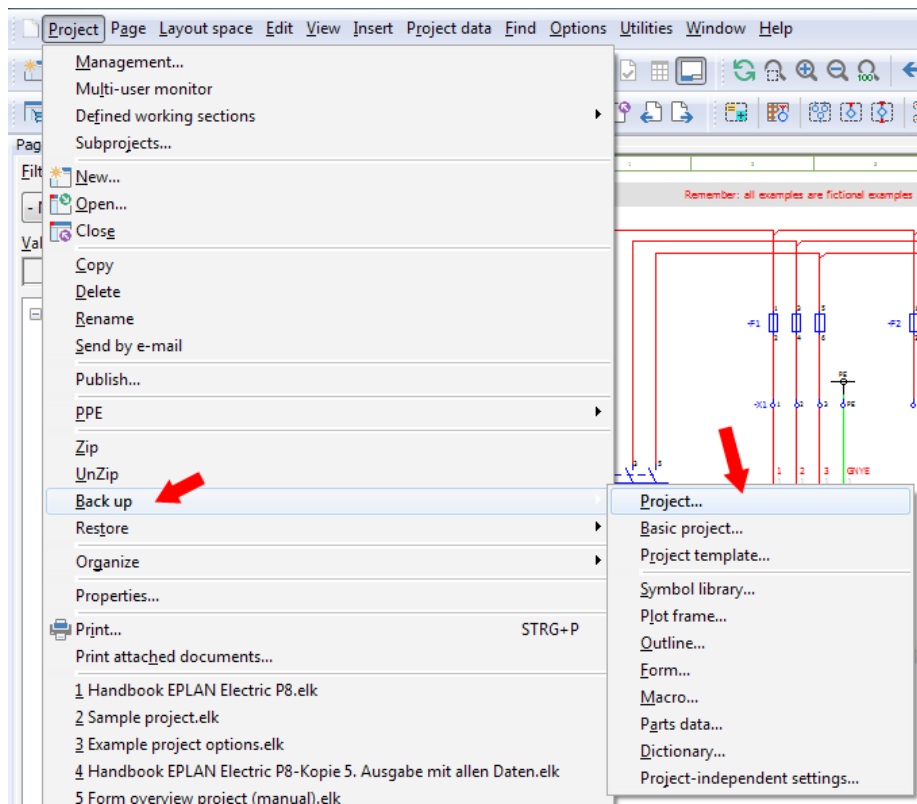


Fig. 9.1
Menu showing backup methods

■ 9.1 Zipping and unzipping of projects

In EPLAN, zipping a project means that the entire project directory is compressed into a single space-saving project file (plus the project information file *ProjectInfo.xml*).

9.1.1 Zip projects

Zippping is started via the **PROJECT/ZIP** menu. After a confirmation prompt, EPLAN immediately zips the project. The *Zip* function itself does not require any special settings.

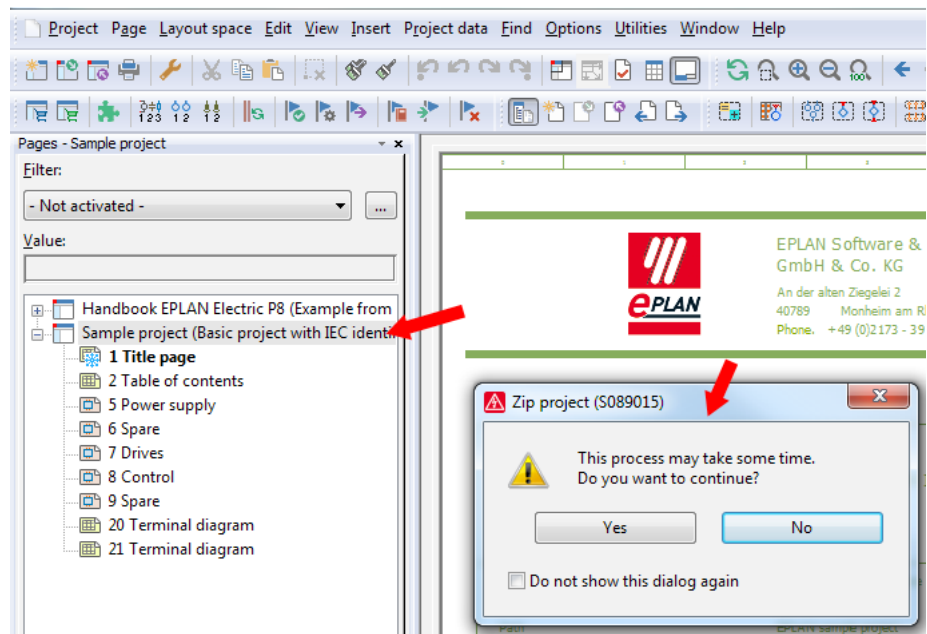


Fig. 9.2 Prompt whether the project is to be zipped

All data in the project directory is zipped. EPLAN leaves only the project properties information (*ProjectInfo.xml*) and the zipped project file (*[Projectname].zw0*) in the project directory and, at a higher level, the project start file *Projectname.elp*.

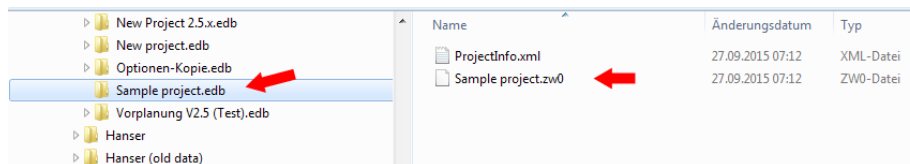


Fig. 9.3 Project data after zipping (packing)



NOTE: Only one project at a time can be zipped via the **PROJECT** menu. Zipped projects are no longer displayed in the page navigator because after being zipped, they are automatically closed and visually removed from the page navigator.

Once EPLAN has completed the **ZIP** function (of the project), it opens a dialog with a message, and the project “disappears” from the page navigator.

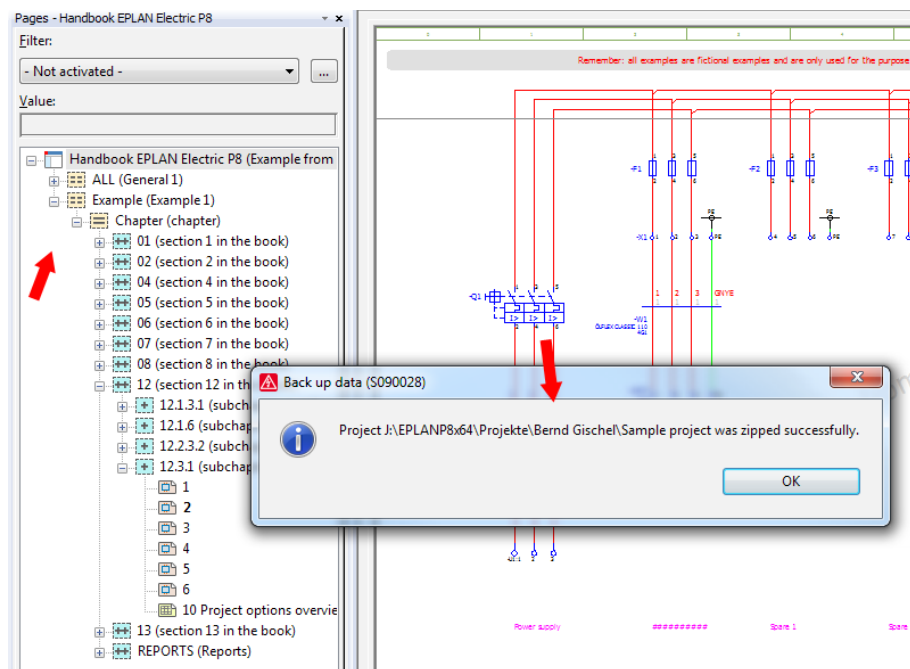


Fig. 9.4 “Project was successfully zipped.” message

9.1.2 Unzip projects

A zipped project can be unzipped for editing in various ways.

One way is to select the project via the **PROJECT/UNZIP** menu and then unzip it by clicking the **OPEN** button. When the project has been unzipped, it will not be displayed in the page navigator. The unzipped project must first be opened via **PROJECT/OPEN**.

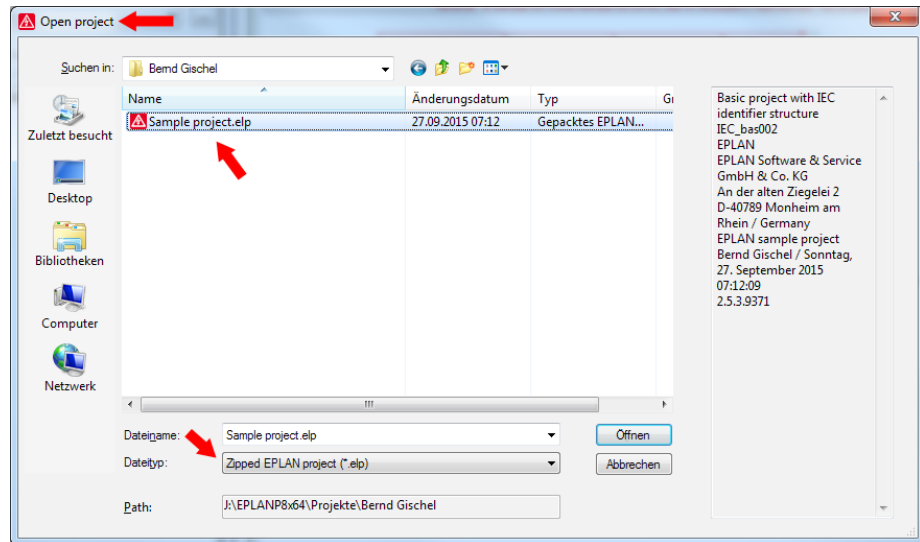


Fig. 9.5 Selecting project for unzipping (unpacking)

Another way is to select the zipped project as usual via **PROJECT/OPEN**. Here, too, the project is unzipped, but unlike in the previous option, it is opened in the page navigator immediately after being unzipped.



NOTE: If zipped (packed) projects are not displayed, the appropriate file type **Packed EPLAN project (*.elp)** must be selected in the dialog.

When a project is unzipped, EPLAN automatically opens the **Restore** dialog. The project is unzipped and then the page navigator is opened. Depending on the unzipping method, the unzipped project is added to any other, possibly open, projects there. The project can immediately be edited.

■ 9.2 Backing up and restoring projects

Compared to zipping a project, the **BACKUP** and **RESTORE** options have many more settings for customizing the data backup and restoration processes.

9.2.1 Back up projects

Unlike the *Zip* function, the *Backup* function is used to file off projects or to be able to sent them to external customers. This is the main difference between these two methods. A zipped project can, of course, also be sent to customers, but it makes more sense to use the *Backup* function for this.

When working on a project, the backup function can also be directly called from the page navigator via the **PROJECT/BACKUP/PROJECT** menu.

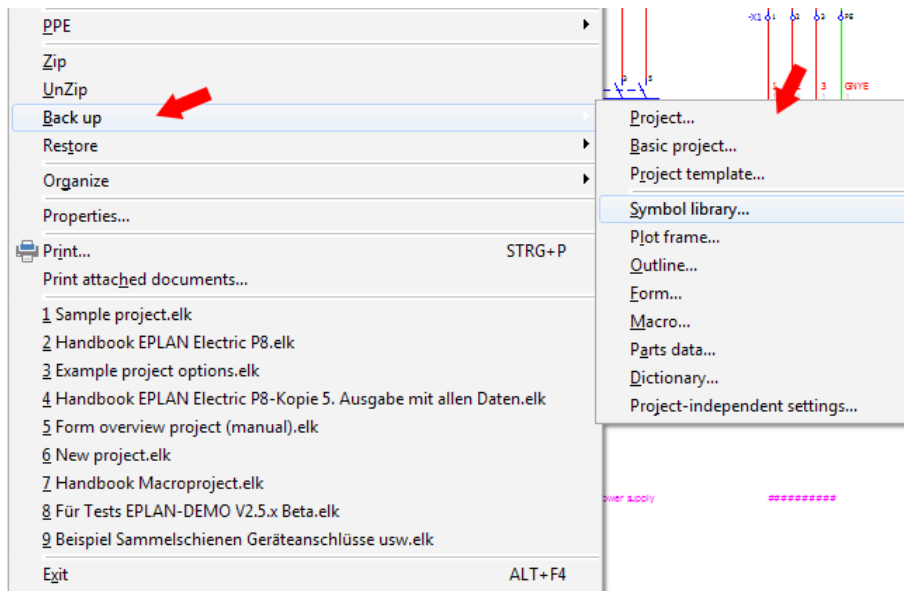


Fig. 9.6 Backup menu and its options

9.2.1.1 Backup options (in the page navigator)

Before starting the **BACKUP** function, you first select the project to be backed up in the page navigator. Now you can back it up via the **PROJECT/BACKUP/PROJECT** menu. EPLAN then displays the **Back up projects** dialog. The dialog allows you to define further settings affecting the backup scope for the project.

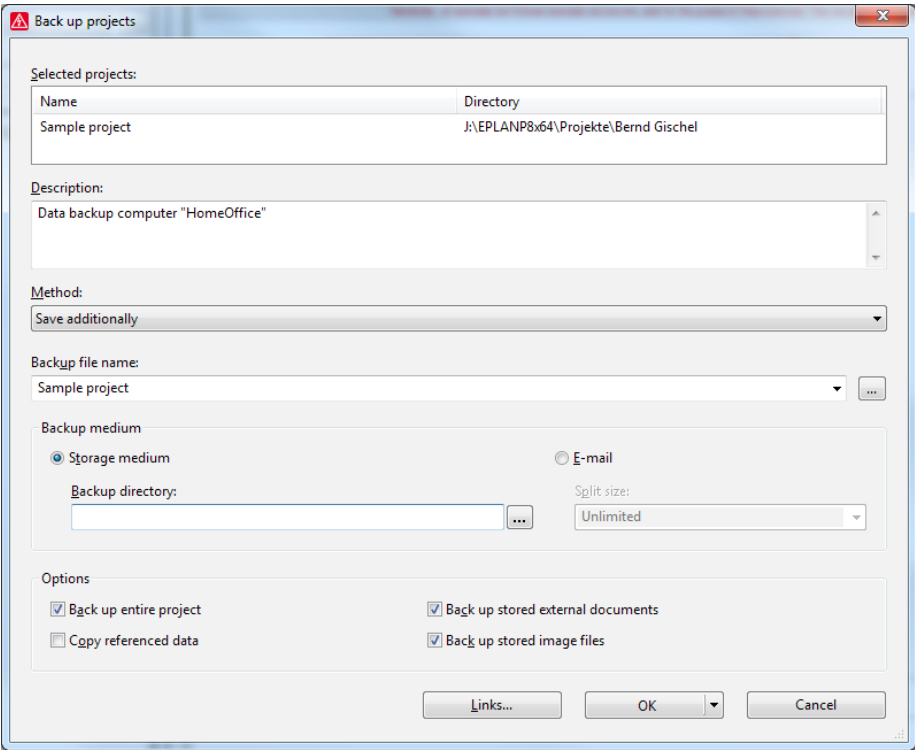


Fig. 9.7
Back up projects
dialog

The **Selected projects** field contains the project name that was selected for backup in the page navigator. EPLAN applies it automatically and, next to it, displays the additional directory information for the project.



NOTE: The *Description* field is informal. This means that you can optionally enter something here. It has no effect on the actual backup process. The description is displayed in the dialog when restoring a project.

The **Method** selection field defines the manner in which the backup is to occur. This will also affect whether or not you can continue to edit the project.
EPLAN provides the user with the following options.

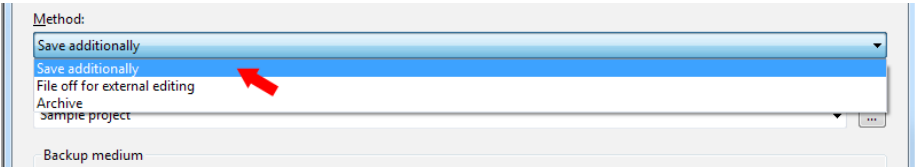


Fig. 9.8
Possible options for
the backup method